CHEMICAL REVIEWS

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Review

From Glycocode to Precision Oncology: Therapeutic Strategies Targeting Aberrant *O*-GalNAc Glycosylation in Cancer

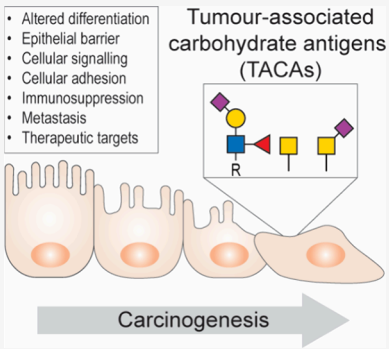
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ABSTRACT: Aberrant O-glycosylation is a prominent feature of cancer and contributes directly to multiple hallmarks of malignant transformation. Cancer-associated O-glycans include both truncated structures that are unique to mucin-type O-glycosylation and more elongated terminal motifs that function as ligands for glycan-binding proteins. Instead of representing a simple binary change from elongated to truncated structures, it is increasingly understood that these glycans decorate the cancer cell surface with a heterogeneous pattern of truncated, elongated, capped, and partially processed glycoforms, which coexist even on individual proteins and within the tumor glycocalyx. Functionally, truncated aberrant O-GalNAc glycans, such as Tn, STn, and T perturb epithelial differentiation, barrier integrity, signaling, and immunosuppression through interactions with glycan binding receptors, such as Siglecs and MGL. In addition, more elongated O-glycans, including Lewis antigens, interact with selectins promoting metastasis and cellular trafficking. Moreover, individual GalNAc-transferase isoenzymes, governing site-specific O-glycosylation, are emerging as context-dependent regulators of oncogenic processes through glycosylation of selected proteins. These observations establish cancer-associated O-glycans as context-dependent molecular drivers of tumor progression and promising therapeutic entry points. In this review, we discuss the expression, regulation, and functions of cancer-associated O-glycans, their roles in glycan-binding protein biology, and emerging therapeutic strategies that may enable a new generation of glycan-directed precision treatments in oncology.



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1. INTRODUCTION

Glycosylation is one of the most abundant and diverse post-translational modifications (PTMs), helping to shape and tune the structure, stability, and functions of proteins and lipids.[1,2] It is therefore not surprising that cancer cells frequently alter glycosylation to influence tumor growth, metastasis, and immune evasion. This makes aberrant glycosylation a common feature of malignant transformation and an important determinant of how tumor cells interact with their surroundings.[3,4]

The cellular glycome comprises different classes of glycoconjugates, including *N*-linked, *C*-linked, and *O*-linked glycans, glycosphingolipids (GSLs), glycosaminoglycans (GAGs), and other glycan-modified structures[5,6] (Figure 1). In cancer, glycosylation undergoes characteristic remodeling, and some of the most cancer-specific and characteristic changes are seen in *O*-GalNAc glycosylation, where expression increases both for short truncated structures, such as Tn (Thomsen-nouveau antigen; GalNAcα1-*O*-Ser/Thr/Tyr), T (Thomsen-Friedenreich antigen; Galβ1,3-GalNAcα1-*O*-Ser/Thr/Tyr), and sialyl-Tn (STn; Neu5Acα2,6-GalNAcα1-*O*-Ser/Thr/Tyr), and for more extended glycans, such as Lewis and sialyl-Lewis antigens. Truncated structures such as Tn, STn, and T antigens are unique to *O*-GalNAc glycans and most often arise from incomplete elongation or altered competition within the *O*-glycosylation pathway, whereas Lewis and sialyl-Lewis structures more often reflect transcriptional changes in elongating and terminal glycosyltransferases.

In this Review, we focus specifically on *O*-GalNAc glycans (hereafter *O*-glycans for simplicity), their alterations in cancer, the effects of these alterations on cellular functions, and how this knowledge can be applied to therapeutic interventions. A central premise is that glycan function depends on interactions with glycan-binding proteins and on cellular context. We therefore describe both how *O*-glycans are remodeled in cancer and how they are recognized by glycan-binding proteins. We further review therapeutic strategies that exploit cancer-associated glycans to selectively target tumor cells or disrupt glycan-mediated functions. Finally, we briefly outline where the field is heading to enable a new generation of glycan-directed precision therapies, checkpoint inhibitors, and targeted interventions.

1.1. Biosynthesis of *O*-GalNAc-Glycans

O-GalNAc glycosylation was historically known as mucin-type *O*-glycosylation because of its abundance on mucins,[7] but it is now recognized as a widespread modification with more than 10,000 individual *O*-GalNAc glycosylation sites present on up to 80% of all proteins traveling through the secretory pathway.[8−17] *O*-GalNAc glycosylation is initiated in the Golgi apparatus by the transfer of N-acetylgalactosamine (GalNAc) to the hydroxyl group of serine, threonine, or occasionally tyrosine residues. In humans, this process is catalyzed by 20 distinct differentially expressed Golgi localized enzymes known as polypeptide GalNAc transferases 1−20 (ppGalNAc-Ts, or simply GalNAc-Ts),[5] which together are responsible for the densely glycosylated patches found in mucin domains and the more isolated O-glycosylation sites on many secreted and membrane proteins.[18−28] Importantly, individual GalNAc-T isoenzymes differ in substrate preference, tissue distribution, and sub-Golgi localization, and both the occupancy and the exact glycosylation site can vary substantially between cell types and physiological states.[28] Following initiation, *O*-GalNAc glycans can be extended into 8 different core structures, of which core 1−4 are the most common[7] (Figure 1). Core 1 and core 2 are the predominant *O*-GalNAc cores in most tissues, whereas core 3 and core 4 structures are more restricted, particularly to gastrointestinal and bronchial epithelia.[29−31] Further elongation includes type 1 LacNAc-containing repeats, usually limited to core 3 structures in the gastrointestinal (GI) tract,[32,33] or the more common type 2 repeats.[34,35] Both the core and repeat structures may then be capped by sialylation, fucosylation, sulfation, generating a highly diverse repertoire of *O*-glycans.[1,7,35−38] For details, please see Figure 1 and the associated legend.

2. SURFACE EXPOSED ABERRANT O-GALNAC GLYCANS ARE A HALLMARK OF CANCER

2.1. Cancer-Associated *O*-Glycans

The broad spectrum of cancer-associated *O*-glycans, including truncated structures such as Tn, STn, and T, as well as elongated and terminal Lewis-type glycans such as sLe^a and sLe^x, are expressed differently between cancer types, between patients, and even differentially within individual tumor lesions. This suggests that distinct glycan structures may serve specialized functions during tumor development and in different tumor regions, such as the stem cell niche or the invasive front.[39−42] Cancer-associated changes in the expression of the initiating GalNAc-T isoenzymes can further reshape O-glycosylation by altering both the selection of glycosylation sites and their occupancy on individual glycoproteins. As such, neither changes in glycosylation sites, nor the shift to the shorter Tn and STn glycans, should be viewed as a simple binary switch. Instead, the cancer associated changes form a heterogeneous landscape of cell-surface exposed glycans, in which truncated, elongated, capped, and partially processed *O*-glycans coexist, even at adjacent glycosylation sites within a single glycoprotein.[40,43] This is relevant for both glycan function and therapeutic targeting, since different glycoforms occupy distinct spatial niches and engage different glycan-binding proteins.[1,28] Recently, a dual-MS approach was developed to analyze the spatial expression of distinct glycoforms of mucin glycopeptides in three colorectal mucinous adenocarcinomas.[44] The study confirmed that different glycoforms stratify regions within each tumor. It will be interesting to see whether the method can be adapted for broader spatial glycoproteomics of tumor tissues in the future.

Across carcinomas, prevalences of surface-exposed Tn and STn generally range from 20−80% (Figure 2) in all major cancer types, although reported estimates vary with antibody specificity and scoring criteria.[37,40,45,46] Expression of Tn and STn often show heterogeneous intratumoral distribution.[47,48] Expression is detectable already in precancerous lesions, maintained across stages I−IV, and often increased in metastases.[40,49,50] Prognostically, STn shows the clearest adverse signal, as

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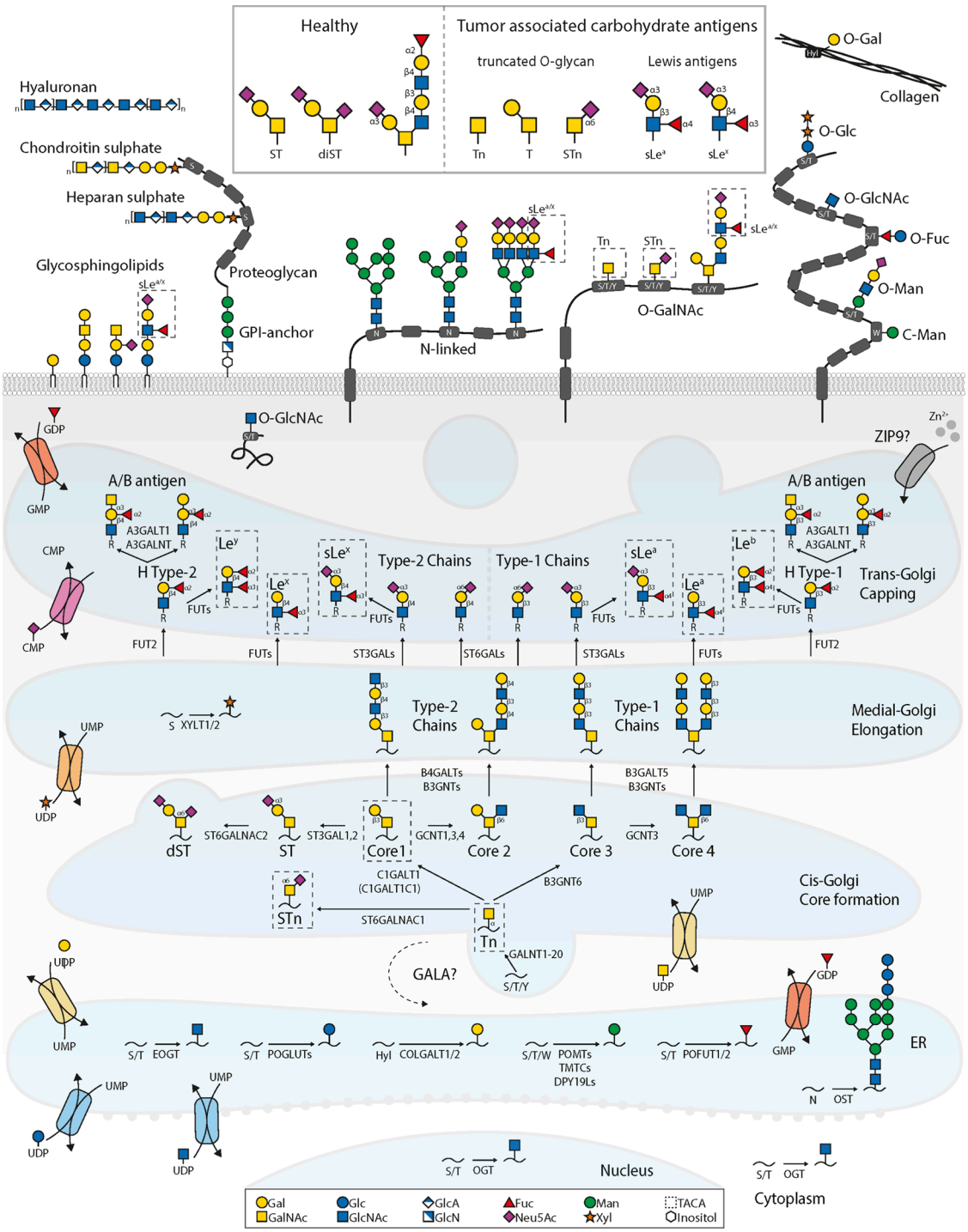


Figure 1. This figure illustrates the glycocalyx and the secretory pathway, showing the various types of glycosylation and glycoconjugates that comprise the cellular glycome and in which subcellular compartments they are initiated. As the focus of this review, the *O*-GalNAc-glycan synthesis is shown in detail. The core 1 (Galβ1−3GalNAcα1-*O*-Ser/Thr/Tyr, also known as the Thomsen-Friedenreich (T) antigen) and core 3 (GlcNAcβ1−3GalNAcα1-*O*-Ser/Thr/Tyr) structures are formed by elongation of the Tn-antigen by the C1GalT1 and C3GnT enzymes, respectively, of which C1GalT1 uniquely relies on the dedicated chaperone COSMC for proper folding, localization, and activity (Springer and Desai 1975; Samuel et al. 1990; Iwai et al. 2002; Ju and Cummings 2002; Wang et al. 2010). The core 1 and 3 structures can be branched into the core 2 (GlcNAcβ1−6(Galβ1−3)GalNAcα-1-*O*-Ser/Thr/Tyr) and core 4 (GlcNAcβ1−6(GlcNAcβ1−3)GalNAcα-1-*O*-Ser/Thr/Tyr) structures, respectively, by the three C2GnT enzymes, of which only C2GnT-2 is able to form the core 4 structure (Bierhuizen and Fukuda 1992; Schwientek et al. 1999; Schwientek et al. 1999; Yeh et al. 1999; Yeh et al. 1999; Schwientek et al. 2000). While C1GalT1, COSMC, and C2GnT-1 and therefore the core 1 and core 2 structures are expressed in most tissues (Stone et al. 2009), C3GnT and C2GnT-2 and therefore the core 3 and core 4 structures are primarily expressed in the gastrointestinal (GI) tract (Iwai et al. 2002; Iwai et al. 2005:3;[29] An et al. 2007; Stone et al. 2009; Arike et al. 2017 Jan 24; Bergstrom et al. 2017). All core structures may undergo elongation and capping, leading to expression of highly diverse structures. Of particular importance for this review, Lewis epitopes on *O*-glycans are generated on terminal type 1 (Galβ1−3GlcNAc) or type 2 (Galβ1−4GlcNAc) lactosamine units on extended core structures (de Haan, 2024). On type 1 chains, FUT3 generates Le^a, whereas prior α1,2-fucosylation by FUT2 yields type 1 H that subsequently can be converted by FUT3 to Le^b (Holgersson and Löfling 2006; Clausen and Hakomori 1989). On type 2 chains, α1,3-fucosylation by FUT3 and related α1,3-fucosyltransferases

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Figure 1. continued

generates Le^x, and additional α1,2-fucosylation by FUT1/2 yields Le^y (Holgersson and Löfling 2006; Clausen and Hakomori 1989). The sialylated Lewis antigens are formed after α2,3-sialylation of these precursor chains followed by fucosylation: sLe^a is produced on type 1 chains mainly through FUT3, whereas sLe^x is produced on type 2 chains through FUT3/4/5/6/7, with FUT7 being particularly important for high-affinity selectin-binding sLe^x on core 2 O-glycans (Ellies et al. 1998, Immunity; Snapp et al. 2001, Blood). Related Lewis termini may also be further modified by sulfation, for example to generate 6-sulfo-sLe^x (Essentials of Glycobiology 2022). Common inherited variation in FUT2 (secretor status) and especially FUT3 strongly influences baseline Lewis expression in the population. Non- or weak-secretor FUT2 genotypes occur in about 20−25% of most populations, whereas inherited loss of FUT3 activity gives rise to Lewis-negative phenotypes in roughly 10−20% overall, but can exceed 20% in some ancestries (Marionneau 2001; Tonini 2020). Several *O*-GalNAc-glycans are upregulated in cancer and are known as tumor-associated carbohydrate antigens (TACAs). These include the truncated Tn- and STn-antigens that are unique to O-GalNAc-glycans and terminal TACA structures, such as $sLe^{a/x}$, that are shared by several glycoconjugates, including glycosphingolipids, *N*-, and *O*-GalNAc glycans.

reported in a meta-analysis based on 11 tissue-based studies estimating an odds ratio for death of 3.26 (95% CI 2.12−5.02) for STn-positive tumors.[11] Moreover, Tn and STn are positively linked to worse survival in gastric cancer.[51] In contrast to carcinomas, Tn and STn are essentially absent from hematological and mesenchymal malignancies and rarely detected in tumors of the nervous system (Tn 5%, STn 1%).[40]

Given the cancer selectivity of Tn and STn there has obviously been great interest in developing robust analytical detection methods to routinely assess their prevalence in tumors. However, this has been difficult, with large variability observed between studies as mentioned above and seen in Figure 2. The variability is mainly explained by differences in sample type, detection reagents, and scoring criteria. Furthermore, in the case of Tn, many historical reagents, such as the lectins VVA and HPA, are not strictly Tn-specific and also bind other GalNAc-containing structures, including blood group A, and Forssman, and IgA1-associated glycans.[52,53] In addition, monoclonal antibodies may differ in fine specificity depending on peptide context, local glycan presentation, and epitope clustering.[40,54] Because many cancer-associated O-glycans also occur as normal biosynthetic intermediates inside cells, studies that do not distinguish true cell-surface expression from intracellular staining may overestimate cancer-specific expression. Reported prevalences should therefore be interpreted primarily as broad indicators of glycan occurrence rather than directly comparable measures of targetable surface expression. Finally, the degree of cancer specificity differs between the Tn-, STn-, and T-antigens. While surface expression of the Tn-antigen is highly cancer specific, the STn-antigen is also expressed in several inflammatory conditions particularly in the GI tract, including gastritis, ulcerative colitis, and Crohn's disease.[45,46] This inflammation-induced expression of STn arises, at least in part, from loss of sialic acid acetylation driven by both microbial activity and host-dependent regulation, including upregulation of sialic-acid O-acetylesterase and downregulation of the 9-O-acetyltransferase CASD1.[45,46,55−58] The lack of cancer specificity is even more pronounced for the T-antigen, that is also found in healthy tissues, including keratinocytes in human skin and hematopoietic stem cells.[3,59,60] Thus, while the T-antigen may be overrepresented in carcinomas, it should not be regarded as a true cancer-specific marker.

Unlike the Tn and STn-antigens, the Lewis antigens are not specific to O-glycans. While there is no doubt that sLe^a- and sLe^x-antigens are overexpressed and involved in cancer progression, it is often unclear which glycoconjugates that carry these epitopes, since most studies rely on antibodies that do not distinguish between glycoconjugate carriers.[61−64,7,65−76,3,43,49,52−54,77−82] Nonetheless, glycoprofiling studies do support the expression of Lewis epitopes on tumor-associated O-glycans, with sLe^a preferentially linked to type 1 chains prominent in gastrointestinal epithelia.[32,33] Importantly, sLe^a- and sLe^x-antigens are not unique to cancer, but also expressed in healthy and inflamed tissue, where they mediate selectin-dependent recruitment of leukocytes to activated endothelium.[83] In cancer, the structures are frequently upregulated and repurposed by tumor cells to support vascular adhesion, dissemination, and metastasis through interactions with selectins[83] as we will discuss below. The expression of Lewis antigens has been described in several carcinomas, including colorectal, pancreatic, biliary, gastric, lung, and bladder cancers (Figure 2).[61−64,84,85] Representative prevalence data show high, but variable prevalences between different cancers and different studies (Figure 2). In pancreatic and colorectal cancer high expression is correlated with lymphatic invasion, metastasis, and poor postoperative survival.[86,87] The related type 2 Lewis structures, particularly Le^y, have also been linked to aggressive behavior in colon and lung cancer.[88,89] Finally, in hematological malignancies, $sLe^{x/a}$ epitopes on mucin-like O-glycoproteins such as PSGL-1, CD43, and CD44 are well-known in AML and multiple myeloma, where they promote E and P-selectin binding, bone marrow homing, and resistance to therapy.[90,91] In addition, host Lewis status may also influence cancer outcome, as seen in pancreas cancer where the Lewis-negative phenotype ($Le^{(a-b-)}$) is associated with poor survival after surgery in PDAC.[3,7,65−76,92−98]

Notably, the sLe^a antigen is currently one of the few glycan-based serum biomarkers with clinical value. Known as CA-19−9, the antigen is used as a standard clinical biomarker for the diagnosis, prognosis, and monitoring of pancreatic ductal adenocarcinomas (PDACs).[99] CA-19−9 has high prognostic value in PDAC, with high levels linked to worse survival, and normalization of CA19−9 values after surgery being associated with longer survival.[100] In addition, low postoperative levels may predict benefit from adjuvant chemotherapy.[101] It is not suitable for screening because its positive predictive value is too low, and some people cannot produce the antigen due to loss of FUT3 function (see legend Figure 1).[102] Although higher cutoff values improve specificity in screening, this severely impacts sensitivity.[102] New combined Lewis-type multimarker panels perform better than CA19−9 alone and support the development of more accurate glycan assays based on combinatorial markers.

In contrast to the Lewis antigens, the use of Tn and STn in biomarker development are limited because glycoproteins carrying short O-glycans are rapidly cleared.[2] To address this, autoantibody profiling against truncated glycans on mucin peptides has been explored as an "immunological mirror" of tumor-associated O-glycosylation.[103,104] Such studies have demonstrated that such antiglycan autoantibodies can be used

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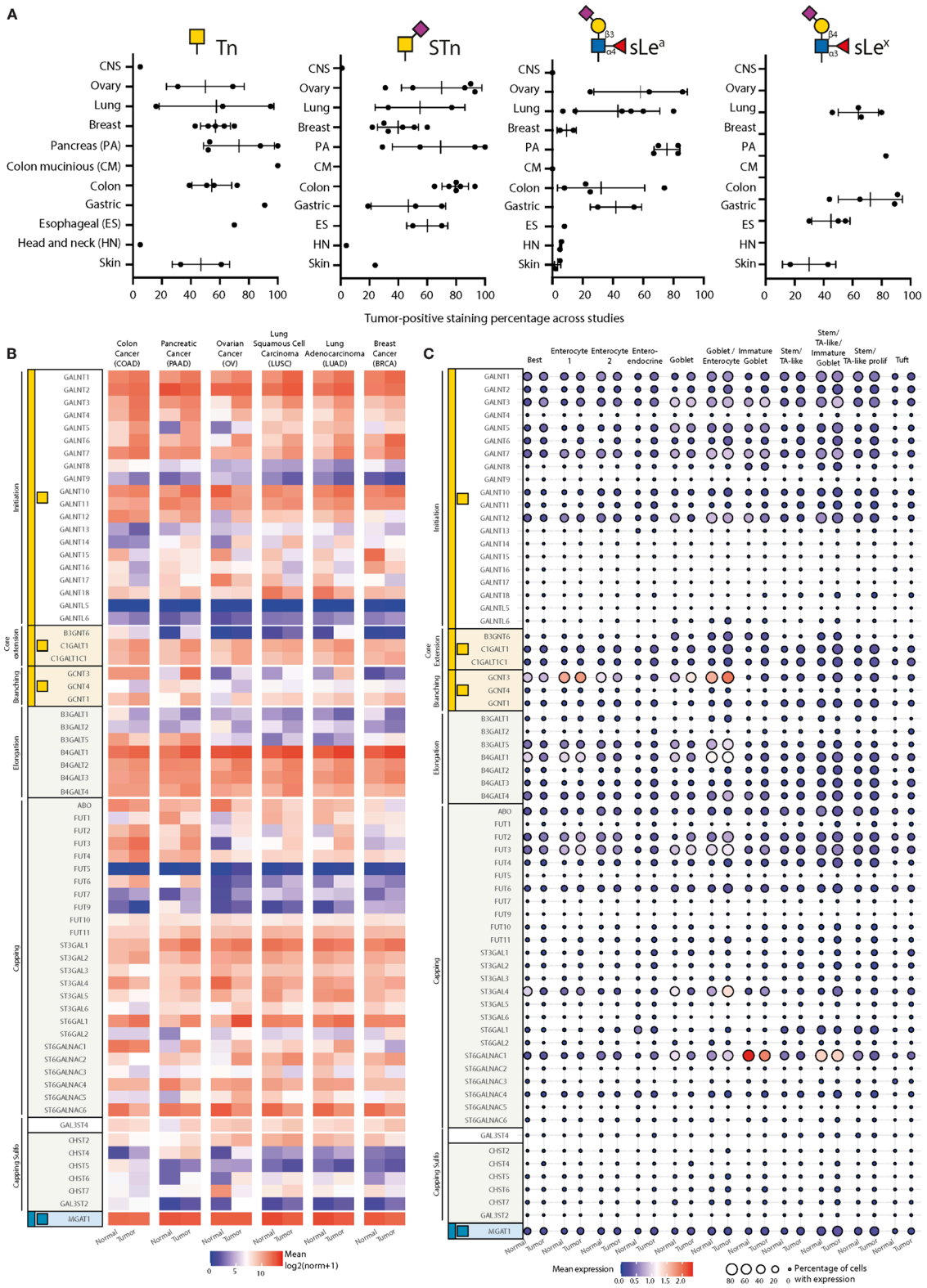


Figure 2. (A) Prevalence of Tn, STn, sLe[a], and sLe[x] expression in cancer tissues reported across multiple studies using monoclonal antibodies, highlighting substantial interstudy and intercancer variability (see Supporting Information, Table 1). (B) Comparing the expression patterns as obtained from the Cancer Genome Atlas (TCGA), for the genes encoding glycosyltransferases that initiate, extend, branch and cap *O*-glycans, the expression levels are different across different tissues, and the trends of dysregulation in cancer (either up- or down-regulation) are also highly heterogeneous. In contrast, in the *N*-glycosylation pathway, the expression levels of MGAT1 are homogeneous. The heterogeneity of expression exhibited by the *O*-glycosylation pathway genes indicates that, specific changes in expression related to cancer may be confounded by the complex populations of cells where expression values are averaged. (C) Decomposing expression data down to the single cell level, single cell transcriptomics was examined from a recent colorectal cancer (CRC) study,[151] and the expression levels between healthy and tumor were plotted. Any changes in

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Figure 2. continued

expression for the GalNAc-transferases are likely to be subtle in each cell type, while tumor-associated changes in the expression level of some glycosyltransferases are more clearly evident (e.g., for GCNT3 and ST3GAL4) among the goblet and enterocyte cells.

as biomarkers for cancer stage and prognosis and even be used to predict the efficacy of different treatment regimens.[105−111] Some colorectal cancer studies showed promising sensitivity with detection of cancers several years prior to clinical diagnosis,[103,112−114] while others using a different detection platform with poor sensitivity did not.[115] This suggests that assay design and epitope presentation are critical for such assays. In parallel, assays measuring cancer-associated mucins as serum biomarkers have also been developed, including shed variants of MUC1 (CA-15-3 and CA-27-29) and MUC16 (CA-125) and the secreted MUC5AC.[116] While CA-15-3 and CA-27-29 are widely used for monitoring of breast cancer[117] and CA-125 is used as a standard serum biomarker for the diagnosis and monitoring of ovarian carcinomas,[118,119] the prognostic value of MUC5AC in PDAC is still being investigated.[120] Of particular interest to this review, investigations of CA-125 glycoforms carrying truncated O-glycans as refined serum biomarkers in ovarian cancer have shown that CA-125-STn provides higher sensitivity in the assessment of tumor load and increased precision in the detection of relapse compared to conventional CA-125,[121] while CA-125-Tn could be a new biomarker for peritoneal dissemination.[122] Looking ahead, it will be interesting to see whether glycoform-specific assays of other shed mucin domains or related *O*-glycoproteins will emerge as novel cancer biomarkers. Potentially, carrier-resolved LC−MS/MS glycoproteomics as well as single molecule glycomic analysis may enable more sensitive, site-specific O-glycoform signatures with improved diagnostic and predictive value.[123,124]

In conclusion, cancer-associated O-glycans comprise a heterogeneous and context-dependent spectrum ranging from truncated structures to more complex Lewis antigens, with substantial variation in cancer specificity and prevalence across cancer types, patients, and even within individual tumors. Accordingly, their biological interpretation and diagnostic and therapeutic exploitation require careful attention to cancer specificity, glycan context, and the current limitations of the detection reagents used to define their expression. It is the hope that improved protocols for analyzing O-glycan structures, including spatial glycoproteomics[44,125−127] will enable us to determine which glycoforms at specific glycosylation sites that are unique to cancer, which are also found in inflammatory tissues, and how their local presentation influences tumor progression.

2.2. Mechanisms Underlying Aberrant *O*-Glycosylation in Cancer

So why do cancer cells display aberrant *O*-glycans? Multiple, coexisting mechanisms have been proposed. These include genetic and epigenetic alterations affecting glycosyltransferase expression and disruption of the structural and functional integrity of the secretory pathway itself.[3,60,128] Although each of these mechanisms may account for aberrant *O*-glycosylation in specific settings, none appears sufficient to explain the full spectrum of glycan alterations observed across tumors. Thus, a single unifying model for the expression of aberrant *O*-glycans in cancer does not seem to be feasible. A useful distinction is that the more complex glycans, such as Lewis structures, are best viewed as inflammation-associated glycoforms and more likely to reflect cytokine-driven inflammatory programs than a general failure of glycan processing in the secretory pathway. For example, as mentioned above, the concurrent overexpression of ST3GAL5 and B3GALT5 has been associated with increased expression of the longer TACA epitopes CA19-9 (sLe^a).[84,129] By contrast, the truncated Tn and STn-antigens seem to be induced by complex changes to the secretory pathway affecting glycosylation elongation. Such changes include protein conformation and neighboring post-translational modifications that influence the accessibility of elongating enzymes. Additional determinants may be Golgi residence time, luminal pH, divalent cation availability, donor-sugar supply, and alternative secretory routing; all of which could alter the duration of exposure of a carrier protein to the elongation machinery. Importantly, these factors are likely to affect isolated and clustered glycosylation sites differently, and may even change the site of glycosylation due to substrate dependent differences in kinetic properties for the individual GalNAc-Ts.[24,26,28] This may help explain why truncated *O*-glycans are frequently observed in mucin-rich domains, where the kinetics of follow-up glycosylation with increased number of acceptors, for example through heavily upregulated mucins, may compromise completion of glycan processing.

The simplest known mechanism for expression of truncated *O*-glycans is the acquisition of deleterious mutations in the chaperone COSMC, required for functional C1GalT1 activity.[130,131] However, this mechanism accounts for only a small minority of cases, estimated around 2−5%,[132] and, more broadly, mutations in genes encoding glycosyltransferases appear to be uncommon and only explain a fraction of tumors with truncated *O*-glycans.[60,133,134] Instead, a transcriptional and epigenetic dysregulation of glycosyltransferases is suggested as a more prevalent mechanism[60,128] (Figure 2). An example is hypermethylation of the COSMC promoter in leukocytes and pancreatic carcinomas expressing the Tn-antigen.[60,135] Other transferase systems might also be regulated through epigenetics, including hypermethylation of the B3GALT5 promoter leading to downregulation of the expression of Type-1 chains on Core-3 *O*-glycans associated with loss of SLe^a in colon cancer as described above.[136−138] Likewise, reduced expression of core 3 structures in colorectal cancers have been attributed to loss of the core 3 synthase, C3GnT.[29] Another example is overexpression of ST6GALNAC1 associated with the premature truncation of O-glycans, through generation of STn-antigen in gastric-, colorectal- and invasive bladder cancers.[139−146] Besides alterations presumed to drive expression of short TACA, downregulation of the core 2 synthase C2GnT-1 with concurrent upregulation of ST3GAL1 in some breast cancers may explain an observed switch from core 2 structures to sialylated core 1 structures,[147−149] a process which has been linked to aberrant expression of the cyclooxygenase-2 (COX-2) enzyme and its metabolite prostaglandin E2 (PGE2).[150]

Other factors, such as altered distribution of glycosyltransferases within the secretory pathway, has been suggested to contribute to the increased expression of truncated *O*-glycans.[152] While most Golgi-resident glycosyltransferases appear to remain confined to Golgi-like structures in cancer cells,[153] the GalNAc-Ts responsible for initiation of *O*-glycan

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biosynthesis have been proposed to undergo Src-dependent retrograde transport to the ER.[154−157] This mechanism, referred to as the GALNT activation (GALA) pathway, has remained controversial.[158] Nevertheless, functional support for the biological consequences of ER-localized GalNAc-T activity comes from studies employing an ER-targeted chimeric GalNAc-T2 construct, which was found to increase the expression of truncated *O*-glycans.[159] Although the physiological relevance of this system can be debated, enforced ER localization of GalNAc-T2 induces hyper-*O*-glycosylation of several proteins, including MMP14 and the ER-resident proteins PDIA4 and calnexin, with consequent enhancement of tumor growth and invasiveness.[160−162] Mechanistically, relocalization of GalNAc-Ts to the ER may expose glycosylation sites that are normally unavailable because of cotranslational or early post-translational folding constraints, or because they are masked by ER-initiated glycosylation events from competing pathways. This concept is exemplified by α-Dystroglycan, as it carries sites that can undergo either *O*-GalNAc or *O*-mannose glycosylation,[163,164] and experimental studies have shown that ER-initiated *O*-mannosylation can directly influence the extent of *O*-GalNAc modification.[165,166] Other mechanisms include release of glycosyltransferases through a signal peptide peptidase-like 3 (SPPL3)-dependent mechanism, in which the Golgi-resident protease SPPL3 cleaves the catalytic ectodomain of these enzymes.[167,168] For example, downregulation of SPPL3 in gliomas has been reported to enhance B3GNT5 activity and increase expression of neolacto-series glycosphingolipids, a phenotype that correlates with patient survival.[169] Although this remains a relatively unexplored mechanism, secretion of GalNAc-T2 has also been shown to depend on SPPL3 activity,[167,168] suggesting that dysregulation of SPPL3 may constitute an additional route by which cancer cells alter *O*-glycosylation.

Altered pH homeostasis, which is an inherent part of the tumor microenvironment, represents an additional mechanism by which glycosyltransferase (GT) function may be dysregulated. Neutralization of Golgi pH in colorectal LS174T cells, achieved with either the proton pump inhibitor bafilomycin or with weak bases, induces relocalization of several GTs, including GalNAc-T2, toward endosomes and the plasma membrane, and mass spectrometry analysis of mucins under these conditions revealed truncation of *O*-glycans.[170] Consistent with this, a separate study in the same cell line showed that bafilomycin increases T antigen expression.[171] In parallel, several GTs have been shown to form homo- and heteromeric complexes through catalytic-domain interactions in a Golgi acidity-dependent manner, such that acidification-defective cancer cells fail to maintain stable complexes and consequently exhibit impaired glycan synthesis.[172] Correspondingly, restoration of acidification normalizes glycan expression.[173,174] At the opposite end of the spectrum, hyperacidification in cystic fibrosis have likewise been reported to reduce sialylation,[175] suggesting a broader relationship between dysregulated pH homeostasis in the secretory pathway and altered glycosylation.[176,177] Importantly, pH regulation within the secretory pathway is likely to be closely coupled to cytoplasmic and extracellular pH, with the Golgi potentially functioning as a "proton sink". Accordingly, manipulation of extracellular pH may have indirect and potentially confounding effects on the Golgi luminal environment. This underscores the importance of measuring pH directly within the Golgi before drawing firm conclusions regarding the relationship between pH and the function of the glycosylation machinery in cancer.

Beyond cell-intrinsic alterations in glycosylation pathways, the inflammatory tumor microenvironment represents an additional and potentially self-reinforcing driver of aberrant glycosylation. It can be speculated that inflammatory signals can reshape glycan composition, while the resulting glycan changes may in turn amplify inflammation and thereby further support tumor development. As discussed above, one example is the expression of nonacetylated STn-antigen in inflammation.[45,46,55−58] In addition, Tn-antigen expression in human gastric cancer has been correlated with both H. pylori infection and inflammatory status.[178] A more direct mechanistic link between inflammatory signaling and truncated O-glycosylation has also been demonstrated in colitis-associated colon cancer, where proinflammatory cytokines were linked to the expression of hypoglycosylated MUC1,[179] and where tumor-resident M2-macrophages were shown to induce ST6GALNAC1 expression in an IL-13/STAT6 mediated pathway.[180] Collectively, these findings highlight the capacity of inflammatory tumor microenvironments to promote truncated *O*-glycan expression through cytokine signaling and immune-cell-mediated regulation.

Finally, genome-wide CRISPR/Cas9 knockout screens identified loss of the ZIP9 zinc transporter as a driver of truncated *O*-glycan expression.[181,182] Given that zinc homeostasis is often dysregulated in tumors, these data suggest a previously unrecognized link between altered Zn^{2+} homeostasis and truncated *O*-glycans.

Taken together, these observations align with earlier deep sequencing studies of glycosyltransferases in pancreatic cancer samples,[60] and more recent surveys,[183] suggesting that altered *O*-glycan expression is explained mainly by transcriptional and epigenetic changes and/or changes in the secretory pathway, rather than by somatic mutations in glycosyltransferases. The diversity of mechanisms at play likely causes the widespread yet heterogeneous expression of truncated *O*-glycans across tumors. Although once debated as bystander changes, increasing evidence suggests that these glycosylation alterations contribute directly to oncogenesis. A central mechanistic question is now, which sites acquire short glycoforms, which sites are elongated, and which local constraints determine these distinct outcomes. It is clear that our current transcriptomic data do not have the capacity to address this question,[13] but maybe this will change, once we have enough information that can link specific scRNA-seq profiles to defined glyco-phenotypes.

2.3. *O*-Glycans Promote Oncogenesis

Both elongated and truncated cancer-associated glycans interact with immune receptors such as selectins, Siglecs, galectins, and MGL,[184] thereby promoting oncogenesis, metastatic spread, and immune escape. These mechanisms are well described and are discussed further below (sections 3.1 and 3.2).

In contrast, we still do not fully understand to which extent cancer-associated *O*-glycans directly influence cells to promote transformation, or the mechanisms by which they do so. Among the most important findings has been the demonstration, in several murine models, that O-glycosylation is essential for epithelial homeostasis. Among these, tissue-specific ablation of C1GalT1 or COSMC in the gastrointestinal tract causes impaired barrier function, spontaneous microbe-dependent colitis, and carcinoma development.[185,186] Similarly, loss of C2GnTs or C3GnT compromises colonic barrier integrity and

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Table 1. Dysregulation of GALNT Isozymes in Cancer Play Diverse Roles to Promote Cancer Progression

Cancer	Model	Effect	Ref
GALNT1			
hepatocellular	human tissue	overexpression correlates with shorter patient survival	(Huang et al., 2015)
	human cell line	knockdown inhibits EGF-induced migration and invasion through EGFR dysregulation	(Huang et al., 2015)
breast	human tissue	overexpression correlates with shorter overall survival	(Milde-Langosch et al., 2015)
	human cell line, CDX	LAMTOR5 promotes expression and ER localization	(Fang et al., 2020)
colorectal	human tissue	overexpressed in cancer cells compared to normal cells, increased in patients with distant metastasis	(Kohsaki et al., 2000)
	human tissue, cell line, CDX	upregulated lncRNA SNHG7 sponges miR-216b to positively regulate GALNT1 expression and promote proliferation and metastasis	(Shan et al., 2018b)
bladder	human tissue, cell line	miR-129 associated with poor prognosis and regulates GALNT1 expression	(Dyrskjøt et al., 2009)
	human tissue	overexpressed in cancer tissues compared to normal tissues	(Ding et al., 2012)
	human cell line, CDX	knockdown reduced proliferation *in vitro* and *in vivo*	(Ding et al., 2012)
	primary cancer cells, PDX	high expression in cancer stem cells maintains stemness through Sonic Hedgehog glycosylation	(Li et al., 2016)
GALNT2			
colorectal	human tissue	overexpressed in cancer cells compared to normal cells, increased in patients with distant metastasis	(Kohsaki et al., 2000)
lung	human tissue	overexpression due to hypomethylation correlates with poor prognosis	(Yu et al., 2021)
oral	human tissue, cell line	overexpressed in invasive front *in vivo* and promotes migration and invasion *in vitro*	(Lin et al., 2014)
glioma	human tissue, cell line, CDX	overexpression correlates with poor prognosis and promotes malignant character through EGFR	(Sun et al., 2019)
gastric	human tissue, cell line, CDX	downregulation correlates with advanced disease stage and shorter survival through MET	(Liu et al., 2016)
hepatocellular	human tissue, cell line, CDX	downregulation correlates with vascular invasion and recurrence through EGFR	(Wu et al., 2011)
neuroblastoma	human tissue, cell line	expression correlates with improved prognosis and suppresses IGF-1R-mediated malignancy	(Ho et al., 2014)
ovarian	human tissue	downregulated compared to healthy tissue	(Sheta et al., 2017)
GALNT3			
colorectal	human tissue	expression correlates with increased patient survival and tumor differentiation and invasion status	(Shibao et al., 2002)
lung	human tissue	downregulation correlates with poor differentiation, tumor stage and recurrence, and poor prognosis	(Gu et al., 2004)
	human tissue	downregulation more frequent in smokers and correlates with K_i-67 labeling and poor prognosis	(Dosaka-Akita et al., 2002)
pancreatic	human tissue	downregulation correlates with poor differentiation and aggressiveness	(Yamamoto et al., 2004)
	human tissue, cell line, CDX	expression promotes cell growth *in vitro* and *in vivo* and protects from apoptosis	(Taniuchi et al., 2011)
	human cell line	overexpressed in cancer stem cells and correlates with cancer stem cell markers, including CD44v6	(Barkeer et al., 2018)
	human tissue, cell line	downregulated in poorly differentiated cancer cells	(Chugh et al., 2016)
gastric	human tissue	expression correlates with differentiation, tumor invasion, and lymph node metastasis	(Ishikawa et al., 2004)
gallbladder	human tissue	overexpressed in stage 2 tumors, increased recurrence and decreased survival in diffuse versus granular localization	(Miyahara et al., 2004)
bile duct	human tissue	expression and diffuse localization at invasive front correlates with lymph node metastasis	(Inoue et al., 2007)
prostate	human tissue	overexpressed together with PSMA, Hepsin, and DD3/PCA3	(Landers et al., 2005)
	human cell line	expression associated with TGF-β-induced EMT	(Freire-de-Lima et al., 2011)
ovarian	human tissue, cell line	overexpressed in high-grade tumors, correlates with shorter survival, and promotes proliferation, migration, and invasion	(Wang et al., 2014)
GALNT5			
gastric	human tissue	reduced expression in tumors correlates with shorter survival	(He et al., 2014)
cholangiocarcinoma	human tissue, cell line	major isoform in tumors, regulates proliferation, migration, and invasion	(Detarya et al., 2020)
hepatocellular	human tissue	expression correlates with poor outcomes	(Deng et al., 2019)

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Table 1. continued

Cancer	Model	Effect	Ref
GALNT5			
pancreatic	human tissue	overexpression correlates with poor prognosis and tumor stage	(Li et al., 2020)
	human cell line	MUC1 negatively regulates GALNT5 expression through interactions with cytoplasmic tail	(Caffrey et al., 2019)
hepatoblastoma	human tissue	upregulated compared to differentiated livers	(Rodrigues et al., 2014)
colorectal	human tissue, cell lines, CDX	downregulation of miR-196b-5p is associated with poor outcome and metastasis through dysregulated GALNT5 expression	(Stiegelbauer et al., 2017)
GALNT6			
breast	human tissue, cell line	expression associated with bone marrow dissemination	(Freire et al., 2006)
	human tissue, cell line	upregulated in tumors, *in vitro* knockdown increases cell adhesion and suppresses cell growth	(Park et al., 2010)
	human tissue, cell line	overexpression correlates with shorter survival, promotes migration and invasion through α2M glycosylation and PI3K/Akt signaling	(Liu et al., 2020)
	human tissue, cell line	expression associated with tumor stage 1 and early carcinogenesis	(Berois et al., 2006b)
	human cell line	expression associated with EMT linked with fibronectin	(Park et al., 2011)
gastric	human tissue	heterogenous tumor expression, associated with venous invasion	(Gomes et al., 2009)
colorectal	human tissue, cell line	upregulated in tumors, expression associated with dysplastic growth	(Lavrsen et al., 2018)
	human tissue, cell line	overexpression correlates with increased survival	(Ubillos et al., 2018)
prostate	human cell line	expression associated with TGF-β-induced EMT	(Freire-de-Lima et al., 2011)
pancreatic	human cell line	expression associated with EMT, knockdown increases E-cadherin and decreases MUC4, EGFR2, and cell viability	(Tarhan et al., 2016)
ovarian	human tissue	overexpression correlates with poor prognosis	(Sheta et al., 2017)
multiple	human cell line	glycosylates and stabilizes GRP78 to suppress apoptosis	(Lin et al., 2017a)
GALNT7			
melanoma	human tissue	upregulated in superficial spreading compared to nodular melanoma	(Rose et al., 2011)
	human tissue, cell line	downregulated by miR-30b/30d to promote metastatic behavior	(Gaziel-Sovran et al., 2011)
hepatocellular	human cell line, CDX	downregulated by miR-17 to promote carcinogenesis, angiogenesis, proliferation, and migration	(Shan et al., 2013)
cervical	human tissue, cell line, CDX	upregulated in tumors, expression negatively correlated with miR-30e which suppresses tumor growth	(Wu et al., 2017)
	human tissue, cell line, CDX	upregulated in tumors, targeted by miR-125a-5p to suppress tumor growth and invasion	(Cao et al., 2020)
	human tissue, cell line	upregulation through downregulation of miR-214 promotes proliferation, migration, and invasion	(Peng et al., 2012)
laryngeal	human tissue, cell line, CDX	targeted by miR-154, which is downregulated in tumors and associated with poor prognosis, to suppress proliferation and migration	(Niu et al., 2018)
colorectal	human tissue, cell line, CDX	upregulated in tumors through negative regulation of miR-34a by lncRNA-SNHG7 to promote proliferation and metastasis	(Li et al., 2018)
	human tissue, cell line	upregulated in tumors through negative regulation of miR-363−3p by lncRNA-CCDC144NL-AS1 to promote growth	(Zhang et al., 2022)
esophageal	human tissue, cell line, CDX	upregulation by decreased expression of negative regulator miR-214 correlates with poor differentiation, invasion, and lymph node metastasis	(Lu et al., 2016)
nasopharyngeal	human tissue, cell line	targeted by miR-494, whose downregulated in tumors is associated with proliferation, migration, and invasion	(Duan et al., 2015)
glioma	human tissue	overexpression correlates with tumor grade, survival rate, possibly through promotion of proliferative and invasive properties	(Hua et al., 2018)
	human tissue, cell line	upregulated in tumors through negative regulation of miR-103a by lncRNA-TP73-AS1 to promote growth	(Wang et al., 2020)
BCP-ALL-MLL-r	primary human cells	upregulated in cancer cells	(Oliveira et al., 2021)
prostate	human tissue	expression may be associated with prognosis in TMPRSS2-ERG subtype tumors	(Kudryavtseva et al., 2019)

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Table 1. continued

Cancer	Model	Effect	Ref
GALNT7			
thyroid	human tissue	expression shows iodine-131 exposure dose-dependency	(Abend et al., 2012)
	human tissue, cell line, CDX	targeted by miR-30b-5p, whose downregulation in tumors is associated with poor clinicopathological features	(Wang et al., 2021)
GALNT8			
breast	human tissue	overexpression correlates with shorter overall survival	(Milde-Langosch et al., 2015)
	human tissue, cell line	expression negatively correlates with prognosis and metastatic potential through regulation of EMT markers	(Huang et al., 2022a)
	human tissue	expression is associated with survival through regulation of ERα expression by BMPR1A glycosylation	(Huang et al., 2022b)
colorectal	human tissue	part of a seven-glycogene prognostic signature in proficient mismatch repair colorectal adenocarcinomas	(Li et al., 2021)
neuroblastoma	human cell line, CDX	expression is promoted by lncRNA-GAU1 through TCEA1 recruitment	(Chai et al., 2018)
GALNT9			
ovarian	human tissue	overexpressed compared to healthy tissue	(Sheta et al., 2017)
neuroblastoma	human tissue, cell line	expression correlates with improved survival and is expressed in primary tumors and S-type cell lines but not metastatic or N-type cell lines	(Berois et al., 2006a)
GALNT10			
hepatocellular	human tissue, cell line	promotes proliferation and inhibits apoptosis, targeted by miR-122 whose downregulation in tumors correlates with venous invasion and poor prognosis	(Wu et al., 2015a)
renal cell	human tissue	overexpression correlates with advanced TNM stage, poor survival, and early recurrence	(Wu et al., 2015b)
	human tissue	overexpression correlates with shorter survival	(Liu et al., 2017)
gastric	human tissue, cell line	overexpression associates with metastasis and poor prognosis through regulation of proliferation and migration	(Xu et al., 2020)
prostate	primary human cells, PDX	genomic alterations reduces disease-free survival	(Lin et al., 2017b)
cholangiocarcinoma	human tissue, cell line	upregulation of lncRNA-DLGAP1-AS2 correlates with poor prognosis, negatively regulates miR-505 which targets GALNT10	(Liu et al., 2021)
ovarian	human tissue, cell line	upregulated epigenetically by BAG3 in platin-resistant tumors to promote cancer stem cells	(Zhao et al., 2021)
	human tissue	overexpression associates with reduced survival, increased infiltration of regulatory T cells, and decreased granzyme B expression by CD8 T cells	(Zhang et al., 2020)
GALNT12			
colorectal	human patients	deleterious germline variants are associated with susceptibility	(Clarke et al., 2012)
		deleterious germline variants are associated with susceptibility	(Evans et al., 2018)
		deleterious germline variants and somatic mutations are associated with susceptibility	(Guda et al., 2009)
GALNT13			
neuroblastoma	human tissue, cell line	marker for bone marrow involvement	(Berois et al., 2006a)
Lewis lung	murine cell line, CDX	upregulated in high metastatic sublines to promote invasion and motility	(Matsumoto et al., 2012)
	murine cell line, CDX	expression leads to trimeric Tn-antigen on syndecan 1 to promote complex formation with integrin α5β1 and MMP9	(Matsumoto et al., 2013)
lung	human tissue	expression correlates with shorter survival	(Nogimori et al., 2016)
breast	human tissue, cell line	overexpression correlates with lymph node metastasis and shorter survival	(Fernandez et al., 2021)
GALNT14			
breast	human tissue	overexpression correlates with shorter overall survival	(Milde-Langosch et al., 2015)
	human cell line	expression associated with multidrug resistance	(Shan et al., 2018a)
	human cell line	overexpression enhances migration, invasion, and proliferation	(Huanna et al., 2015)
	human tissue, cell line, CDX	expression promotes lung-specific metastasis and modulates self-renewal	(Song et al., 2016)
lung	human tissue	overexpression due to hypomethylation correlates with poor prognosis	(Yu et al., 2021)
	human patients	nonsignificant trend toward favorable survival in patients with overexpression receiving dulanermin	(Soria et al., 2011)
	human tissue, cell line, CDX	expression promotes metastasis through Wnt-dependent HOXB9 upregulation	(Kwon et al., 2015)
	human tissue, cell line, CDX	expression drives metastasis likely through regulation of TGF-β-dependent metastatic genes	(Kwon et al., 2020)

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Table 1. continued

Cancer	Model	Effect	Ref
GALNT14			
ovarian	human tissue	overexpression correlates with poor prognosis	(Sheta et al., 2017)
	human cell line	upregulated by BORIS to promote migration and invasion	(Hillman et al., 2019)
multiple	human cell line	overexpressed in various cancer cell lines and regulates Apo2L/TRAIL sensitivity	(Wagner et al., 2007)

increases susceptibility to colitis[187,188] and adenocarcinoma formation.[187] In addition, the combined loss of core 1 and core 3 derived *O*-glycans in intestinal epithelial cells produces an even more severe phenotype, with spontaneous colitis throughout the colon and evidence of increased proteolytic vulnerability of the mucin barrier.[189] Moreover, in the stomach, epithelial-specific ablation of COSMC likewise induces spontaneous gastritis-associated cancer accompanied by dysregulated expression of MUC1 and MUC5AC and altered gastric acid homeostasis.[178] Finally, mice with duodenum-specific ablation of C1GalT1 and C3GnT developed spontaneous tumors with age, again associated with a compromised mucus barrier, but without overt signs of infection.[190]

These findings show that elongated *O*-glycans are required for mucosal barrier integrity in the GI tract, and that truncation of *O*-glycans promotes inflammatory conditions predisposing to carcinoma formation. Mechanistically, and mostly relevant for the effects observed in the GI tract, this likely reflects the key importance of elongated *O*-glycans for mucin organization, supramolecular organization, and susceptibility to proteolysis. Studies of MUC2 have shown that site-specific *O*-glycosylation can shield the mucin polypeptide backbone from protease attack, and recent structural analyses of mucinases have demonstrated that cleavage is strongly governed by local glycan density and glycan structure.[191−194] Thus, truncation of *O*-glycans is expected to weaken the protective properties of the mucin layer by altering mucin conformation, reducing resistance to proteolysis, and possibly by compromising polymer organization. Consistent with this view, studies using synthetic mucin analogues indicate that glycan composition can directly influence peptide-backbone conformation and degradability,[195,196] suggesting that inflammation and cancer-associated glycoforms alter properties of the mucin layer itself. An additional level of regulation may arise from altered interactions with glycan-binding proteins that contribute to barrier assembly, such as trefoil factors acting as divalent lectins for GlcNAc-α1,4-Gal-capped mucin glycans cross-linking mucin polymers through avidity-dependent interactions.[197,198] Thus, aberrant *O*-glycosylation may promote tumorigenesis by reshaping the physical and biochemical properties of mucins in ways that favor proteolytic remodeling, inflammation, and tumor progression.

Although murine models have provided major insight into how O-glycans regulate gastrointestinal epithelial homeostasis, it is difficult to separate barrier- and inflammation-driven effects from cell-intrinsic consequences of truncated *O*-glycans. Since many phenotypes may be influenced by immune cells or microbes in the local microenvironment, the use of 3D human organoid and organotypic models is an alternative way to examine the endogenous effects on tissue homeostasis.[60,199] In genetically engineered epithelial tissues, loss of elongated *O*-glycans reduced cell−cell adhesion, delayed differentiation, increased proliferation, and promoted invasive growth in vitro and after xenotransplantation.[60] Integrated glycoproteomic and phosphoproteomic analyses suggested that *O*-glycan truncation directly disrupts adhesion and downstream signaling.[60] These findings support a cell-intrinsic role for elongated *O*-glycans in maintaining tissue architecture, independent of microbes or immune-cell infiltration.[60,199] Notably, in 3D models based on primary immortalized skin keratinocyte, loss of C1GalT1 or COSMC also reduced cell−cell adhesion but did not induce invasion, supporting the idea that truncated *O*-glycans act as codrivers rather than sole drivers of cancer.[199] This is consistent with murine pancreatic cancer models, where loss of elongated *O*-glycans combined with Kras and Trp53 mutations promoted more aggressive, invasive tumors.[200]

Collectively, these findings show that *O*-glycan truncation can promote oncogenesis through both extrinsic and cell-intrinsic mechanisms. It weakens barrier function and mucin organization, increases susceptibility to inflammation and proteolytic remodeling, and directly disrupts adhesion, differentiation, and signaling. Truncated *O*-glycans therefore appear to act as oncogenic codrivers, although their expression alone is often insufficient for full malignant transformation and instead cooperates with mutations, inflammation, and the local microenvironment.

2.4. Individual GalNAc-T Isoenzymes Emerge As an Oncogenic Driver

While the preceding sections considered the functions of elongated and truncated *O*-glycans as broader glycan classes, an additional level of regulation resides in the site-specific initiation of *O*-GalNAc glycosylation by individual GalNAc-T isoenzymes. The 20 distinct individual GalNAc-T isoenzymes have overlapping but specific substrate preferences, and most family members use a C-terminal lectin domain to recognize previously glycosylated acceptors and thereby support long-range, context-dependent site selection.[24,28,201] A subset, including GalNAc-T7 and GalNAc-T10, instead functions as short-range follow-up enzymes that preferentially glycosylate sites adjacent to pre-existing GalNAc residues.[22,26] Accordingly, altered expression of individual GalNAc-Ts is unlikely to cause a simple global change in the cell-surface *O*-glycome, but rather to reshape site-specific glycosylation according to peptide context, prior glycosylation, and tissue-specific substrate expression.[28,201] Therefore, it is also expected that changes in GalNAc-Ts are linked to multiple processes associated with cancer progression, including cellular signaling, remodeling of adhesive interfaces, epithelial plasticity, and metastatic dissemination.[24,28,202] As summarized in Figure 2 and Table 1, GALNT1, GALNT6, GALNT10, and GALNT14 are most often reported as upregulated in aggressive epithelial cancers, whereas GALNT2, GALNT3, and GALNT5 show bidirectional behavior across tumor types. GALNT7 is emerging as another frequently upregulated isoenzyme, while evidence for GALNT8 and GALNT9 remains limited and more context dependent. By contrast, GALNT12 stands apart in being linked

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most strongly to colorectal cancer susceptibility through deleterious germline or somatic variants rather than recurrent overexpression.

Cancer-associated changes in GalNAc-T expression are suggested to affect canonical signaling, however, in most cases this is more an association than a mechanistic link.[28,202,203] GALNT1, for example, is commonly upregulated in hepatocellular, breast, colorectal, and bladder carcinomas, where higher expression has been associated with poor prognosis, proliferation, and metastasis.[202,204,205] Among the proposed mechanisms is maintenance of stemness, possibly through glycosylation of Sonic Hedgehog as seen in bladder carcinoma,[204] whereas studies in hepatocellular carcinoma suggest that GALNT1 influences EGFR mediated effects on migration and invasion.[205] In both cases the mechanisms have not been proven. Moreover, opposite effects are often reported in different tissues clearly suggesting context dependent functions. For example, GALNT2 is overexpressed in colorectal, oral, lung, and glioma settings, yet downregulated in gastric, hepatocellular, and ovarian cancer, and linked to various not-mechanistically defined changes in EGFR, MET, and IGF1R signaling.[206,207] Again, these connections are mostly through associations, and without identification of a clear mechanism. This makes it difficult to draw simple molecular explanations and instead suggest that the effect of altered GalNAc-T expression on signaling is highly dependent on tissue context and substrate repertoire.

A few exceptions exist where the molecular mechanism behind the site-specific glycosylation and receptor function is beginning to emerge. One example is GALNT14. Early studies identified GALNT14 expression as a predictor of sensitivity to Apo2L/TRAIL and DR5-agonist-based therapies, especially in pancreatic cancer, NSCLC, and melanoma.[208] The studies suggested that GalNAc-T14-dependent glycosylation enhanced clustering of DR4/DR5 and recruited and activated caspase-8 to the Death-Inducing Signaling Complex.[208] Although the specific effect of GalNAc-T14 can be questioned, this model was later strengthened by showing that normal extended *O*-glycans on DR4 and DR5 promote receptor stability and oligomerization and thereby enhance TRAIL-induced apoptosis, whereas truncation to Tn/STn reduces apoptotic responsiveness.[209] Together, these findings suggest that one important way in which altered GalNAc-T expression influences tumor-cell survival is through site specific receptor glycosylation, probably through the concerted action of elongated glycans and glycan binding proteins.[208,209]

Cell−cell and cell−matrix interactions are also suggested to be regulated by GalNAc-Ts, although in many cases this is difficult to distinguish from their effects on receptor signaling.[28] Developmental studies of the *Drosophila* ortholog of human GalNAc-T1, pgant3, first established a role for GalNAc-T activity in integrin-mediated adhesion.[210] Other findings have demonstrated how GALNT2 contributes to regulation of adhesive interfaces, and although not yet fully verified at the substrate level, the function may be mediated through site-specific glycosylation of proteins in adherens junctions, tight junctions, desmosomes, and hemidesmosomes.[13,211] These observations support the view that site-specific *O*-glycosylation can directly modulate cell−cell and cell−matrix adhesion, and that altered occupancy at defined sites may facilitate local invasion. Consistent with this idea, GALNT2 overexpression in colorectal, lung, and oral carcinomas, as well as glioma, has been associated with metastasis and poor prognosis, particularly at the invasive front of tumors.[206,212−214] More broadly, the available evidence suggests that GalNAc-T-dependent regulation of adhesive interfaces may represent an important oncogenic mechanism, but one that is likely mediated through selected glycoprotein targets rather than through a uniform change in all cell-surface glycans.

GalNAc-Ts also regulate epithelial differentiation programs, although such effects are likely to result from effects on signaling and adhesive programs. The clearest examples involve GALNT3 and GALNT6. GALNT3 supports epithelial differentiation, as shown in human skin,[13,211] and its loss promotes epithelial-to-mesenchymal transition (EMT) in mammary epithelial cells and trophoblast stem cells.[13,211,215] In human cancers, down-regulation of GALNT3 has been linked to poor differentiation and aggressive behavior in lung and pancreatic cancer, whereas overexpression has been reported in ovarian, gallbladder, bile duct, and prostate cancers, as well as in stem-like pancreatic subpopulations. By contrast, increased GALNT6 expression has more consistently been associated with dysplastic growth and EMT,[216−219] and has been reported to promote dysplasia, inhibit differentiation, and correlate with venous invasion and poor prognosis in colon, gastric, pancreatic, ovarian, and breast carcinomas.[216,217] In colonocytes and in mammary epithelial cells, enforced GALNT6 expression disrupted differentiation programs and acinar morphogenesis, respectively.[217,220] In connection with EMT formation it was further speculated that GalNAc-T3 and -T6 specific glycosylation of the IIICS region of fibronectin generated a functionally distinct EMT-associated fibronectin glycoform linked to invasive behavior.[218,219,221−231]

A further important layer of regulation of cancer associated GalNAc-Ts is through noncoding RNA.[232] One example is GALNT7 differentially regulated in several carcinomas, including cervical, colorectal, esophageal, glioma, thyroid, and castration-resistant prostate cancer,[145,233,234] where multiple miRNAs and lncRNA are described to control the expression of GALNT7. In melanoma, the metastatic miR-30b/d program suppresses GALNT7 and GALNT1, while miR-17 represses GALNT7 in hepatocellular carcinoma. GALNT14 is likewise subject to post-transcriptional and epigenetic regulation, both by miR-125a, promoter hypomethylation, BORIS-dependent transcription, and the clinically relevant rs9679162 genotype in hepatocellular carcinoma.[235,236] These observations suggest that the expression of GalNAc-Ts is part of a larger transcriptional network, possibly coordinating appropriate substrate expression with the relevant enzymes. Such post-transcriptional regulation is consistent with the miRNA framework developed by Mahal and colleagues, which proposes that disease-relevant miRNA programs can reveal functionally important glycogenes within broader signaling networks.[237,238]

Finally, metastatic dissemination has also been linked to selected GalNAc-Ts. Among the clearest examples are GALNT11, GALNT13, and, in specific settings, GALNT14. Following xenografting of a human neuroblastoma cell line, GALNT11 and GALNT13 were selectively expressed in metastatic bone marrow lesions, whereas GALNT9 was restricted to the primary tumor.[239] In agreement with these findings, GALNT13 has been shown to promote dissemination of lung cancer cells, possibly through modification of syndecan-1 with a trimeric Tn-antigen, thereby facilitating syndecan−integrin−MMP9 complex formation and enhancing motility and extracellular matrix degradation.[240,241] These results are consistent with the association of GALNT13 expression with metastasis and poor clinical outcome in lung cancer, neuro-

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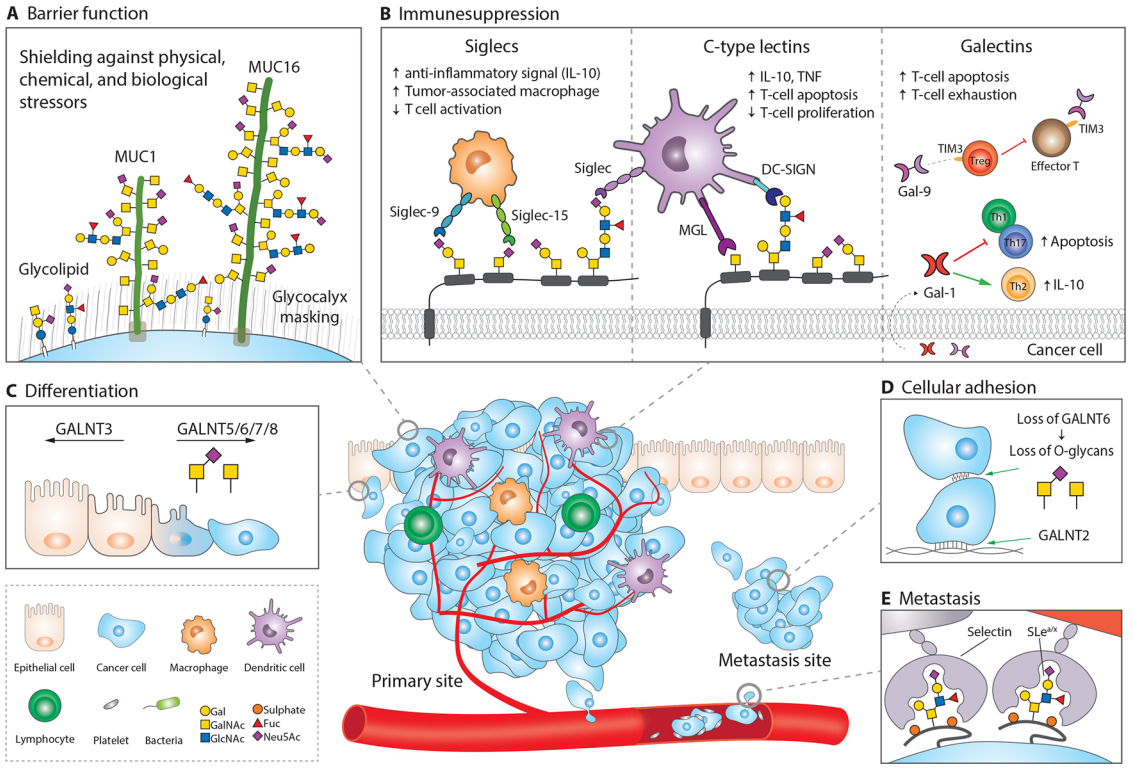


Figure 3. *O*-GalNAc glycosylation directly regulates cancer-associated processes This figure illustrates the various processes linked to oncogenesis and cancer progression that *O*-GalNAc glycosylation has been demonstrated to regulate. (A) Barrier function. O-GalNAc glycosylation of mucins, including MUC1 and MUC16, together with glycolipids and glycocalyx masking, maintains epithelial barrier integrity in the gastrointestinal tract by shielding against physical, chemical, and biological stressors. (B) Immunosuppression. Cancer-associated O-GalNAc glycans engage three major classes of glycan-binding proteins to create an immunosuppressive tumor microenvironment. Siglecs (e.g., Siglec-9, Siglec-15) promote anti-inflammatory signaling (IL-10), tumor-associated macrophage polarization, and suppression of T cell activation. C-type lectins, including MGL and DC-SIGN, induce IL-10 and TNF production, T-cell apoptosis, and inhibition of T-cell proliferation. Galectins (Gal-1 and Gal-9) promote T-cell apoptosis and exhaustion, with Gal-1 driving Th2/Treg skewing with increased IL-10 and Gal-9 engaging TIM-3 on effector T cells and Tregs. (C) Differentiation. GALNT3 maintains epithelial differentiation, whereas overexpression of GALNT5, -6, -7, and -8 are associated with delayed differentiation and promotion of epithelial−mesenchymal transition. (D) Cellular adhesion. GALNT2 promotes adhesion to the extracellular matrix. Conversely, the loss of GALNT6 and consequent loss of its specific O-glycans promote tight cell-to-cell adhesion by reversing epithelial-mesenchymal transition. (E) Metastasis. Upregulation of $sLe^{a/x}$ structures on cancer cells facilitates metastatic dissemination through selectin-mediated adhesion to the vascular endothelium at distant sites. Monosaccharide symbols follow the Symbol Nomenclature for Glycans (SNFG).

blastoma, and metastatic breast carcinoma.[239,242−244] By contrast, GALNT12 remains a special case, since its strongest link is not to metastatic dissemination but to colorectal cancer susceptibility, through inactivating germline and somatic variants whose functional consequences remain incompletely resolved.[245−248]

Together, these findings show that individual GalNAc-Ts may regulate multiple cancer hallmarks through glycosylation of selected substrates rather than broad changes in cell-surface glycans. A major priority for the field is therefore to define and integrate determination of isoform-specific glycosylation sites, expression data such as scRNaseq, and functional perturbation studies to define the cancer-specific consequences of altered GalNAc-T expression. In this regard, new *O*-glycoproteomic tools will play a critical role. For example, since GalNAc analogue, GalNAz, was first introduced for biorthogonal labeling and later used for global glycoproteomic analysis, the recent development of the "bump-hole" chemical reporter system and the Bio-Orthogonal Cell line-specific Tagging of Glycoproteins (BOCTAG) strategy have allowed for identification of substrates specific for individual GalNAc-Ts in living cells with largely intact biosynthetic pathways.[14,249−253] Similarly, the utilization of genetic engineering to develop the SimpleCell platform and genetic knockouts of individual GalNAc-Ts in tissue-forming cell lines have allowed identification of unique and biologically significant glycosylation sites,[8−11,13] Together, these advancements now enable analysis of the GalNAc-T-dependent "glyco-barcode" and its effects on proteolysis, protein conformation, and glycan-mediated interactions in cancer, in a highly site- and context-dependent manner.[12,28,254,255]

3. EFFECTS OF *O*-GALNAC GLYCANS ON CANCER PROGRESSION THROUGH GLYCAN-BINDING PROTEINS

The communication between the cancer-associated *O*-glycans of tumors and their microenvironment involve multiple classes of glycan-binding proteins, including selectins, Siglecs, MGL, DC-SIGN, and galectins,[83,184,256,257] playing a major role in leukocyte trafficking, vascular adhesion, and immune evasion (Figure 3). Both alterations in glycan density, glycan site, and glycan structures could reshape such interactions with importance for the communication between tumor cells, stromal compartments, the vasculature, and the immune system.

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3.1. Selectins Recognize Elongated Cancer-Associated *O*-GalNAc Glycans to Promote Metastasis

While the expression of truncated *O*-glycans contribute to altered differentiation and adhesion during tumorigenesis, elongated glycans carrying terminal sLe$^{a/x}$ structures primarily increase the metastatic potential of cancer cells through interaction with selectins. As briefly mentioned above, the sLe^a- and sLe^x-antigens are classically known in leukocyte trafficking through their interactions with selectins.[83] The minimal recognition epitope of all selectins is the sLe^x structure, but sLe^a and other epitopes may also be recognized and modulate binding strength.[258−262] In physiological conditions, sLe^x-antigens expressed by leukocytes, typically on core 2 *O*-glycans, are recognized by E- and P-selectins expressed by activated endothelial cells in the blood vessels of inflamed tissues to secure leukocyte recruitment.[83,188,263−266] In contrast, L-selectin is expressed mainly in resting immune cells to allow their trafficking to the lymph nodes through interactions with selectin ligands expressed by the high endothelial venules (HEVs).[267] Cancer cells hijack this mechanism to promote metastasis. In leukemia and lymphomas cancer cells have been shown to express L-selectin promoting lymph node infiltration.[268,269] Conversely, as described above, carcinoma cells frequently express sLe^a and sLe^x, often localized on core 2 structures, and use these glycans to adhere to the vascular endothelium through selectin interactions.[64,67,270−273] Consistent with this, expression of sLe^a and sLe^x correlates with invasion of lung cancer into blood vessels[68,69] and dissemination of colon cancer.[71,274−276] In contrast, downregulation of sLe^a and sLe^x decrease the metastatic potential of colon cancer cells,[277−279] and pulmonary metastasis in mice can be inhibited by sLe^a mimetics and knock out of E-selectin.[280]

Additional studies further support the importance of selectin ligands in cancer progression, as partly described above in section 2. Upregulation of sLe^a-antigen in mice promotes severe pancreatitis with hyperactivation of EGFR signaling, and in combination with mutation in the *Kras* system induces pancreatic tumorigenesis.[85] Leukemic cells also hijack the constitutive expression of E-selectin in the bone marrow to metastasise;[281] a mechanism usually utilized by hematopoietic stem cells for homing.[282−284] Furthermore, sLe^a and sLe^x are known to be involved in the formation of emboli consisting of cancer cells and platelets through platelet-expressed P-selectin, promoting metastasis through the capture of emboli by the endothelia.[285−288] Platelet recruitment and activation have also been shown to be involved in angiogenesis in tumors.[289−291] For example, P-selectin glycoprotein ligand (PSGL-1) on cancer cells was identified as phagocytosis checkpoint that enables hematological malignancies escape from macrophage clearance.[228] More broadly, expression of sLe^a and sLe^x across multiple tumor types correlates with poor prognosis and increased invasiveness.[7,65−76]

3.2. Siglecs Recognize Clustered *O*-GalNAc Glycan SAMPs on Mucin Domains to Serve As Immune Checkpoints

Another prominent feature of cancer-associated *O*-glycans is their ability to mediate immunosuppression through interactions with glycan-binding proteins (GBPs), a role that is particularly evident in their engagement of Siglecs. The concept of glycan structures function as tolerance-promoting self-associated molecular patterns (SAMPs) as a parallel to the pathogen- and danger-associated molecular patterns (PAMPs and DAMPs, respectively) was originally proposed by Ajit Varki.[292] Consistent with this, immune cells express a broad repertoire of GBPs that are able to modulate immune activation.[184,256,257] These interactions rely not only on glycan structures but also on protein carrier and density. Heavily glycosylated mucins, such as MUC1, are often overexpressed by cancer cells,[293] allowing cancer cells to form dense clusters of *O*-glycans on their cell surface that may function as SAMPs and promote immunosuppression.[28] Indeed, clustered sialoglycans or Tn-antigen have been shown to modulate interactions with Siglecs,[294,295] other types of GBPs such as the MGL receptor,[296] and microbial adhesins.[297−299] Sialoglycans are broadly upregulated in cancers and are now well recognized as mediators of tumor associated immunomodulation.[300,301] In support of this view, genetic ablation of CMAS in a murine tumor model, resulting in global desialylation, restored antitumor immunity and increased infiltration of lymphoid cells, particularly NK cells and $\gamma\delta$ T cells, to levels exceeding those achieved by anti-PD-L1 therapy.[302] Additionally, a recent study utilizing a genome-wide CRISPR knockout screening found that sialylated *O*-glycans on CD43 was a main driver of inhibition of antibody-dependent and antibody-independent cellular phagocytosis (ADCP and AICP, respectively) of cancer cells, highlighting the cancer-protective and immunomodulatory role of sialoglycans.[303] Although the relative contribution of *O*-glycans can be difficult to distinguish from other glycoconjugates, Siglec-4, 5, 7, 9, 10, 14, and 15 have all been shown to recognize sialylated *O*-glycans in cancer.[294,304]

The 14 human Siglecs are endocytotic receptors that may contain immunoreceptor tyrosine-based inhibitory motifs (ITIMs) or associate with proteins bearing immunoreceptor tyrosine-based activation motifs (ITAMs) to transmit anti- or pro-inflammatory signaling.[305,306] Because many Siglecs exert immunosuppressive functions they are increasingly regarded as immune checkpoints, and tumor-associated hypersialylation is thought to represent a mechanism by which cancers engage Siglecs to dampen immune activation.[256,306,307] Among these receptors, Siglec-7 has been shown to inhibit NK-cell mediated cytotoxicity of cancer cells[308] and promote monocytic differentiation toward tolerogenic tumor-associated macrophages (TAMs) in pancreatic cancer.[309] Siglec-7 is primarily expressed by NK cells, monocytes, macrophages, and dendritic cells, and appears to preferentially bind dST and STn structures, whose recognition was markedly enhanced by 6-*O*-sulphation by CHST1.[294,308,310−312] Studies using overexpression of mucins or mucin-like domains found that Siglec-7 differentially recognizes distinct mucin rich domains carrying dST or STn,[294] indicating that binding not only depend on the specific glycan but also on local glycosylation pattern and presentation, including their density. These findings suggest that Siglec-7 can recognize not only clusters of ST on glycoproteins in healthy tissues, but also clusters of STn characteristic of cancer and inflammation.

Siglec-15 displays a similar recognition pattern, and also binds clustered STn on mucins, likewise with its interaction enhanced by 6-*O*-sulphation by CHST1.[294,312] In contrast to Siglec-7, Siglec-15 only recognizes the α2−6 linkage of STn and not of dST.[294] Although, Siglec-15 lacks an ITIM, it has been shown to possess anti-inflammatory properties,[313] and is upregulated in tumor-infiltrating myeloid cells promoting immunosuppression of antigen-specific T cells *in vitro* and *in vivo*.[313] Due to its preferential recognition of sulfated STn, we can speculate that the physiological role of Siglec-15 is related to the control of adverse immune responses toward the aberrant sulfated STn

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found occasionally in the colon. Like Siglec-7 and Siglec-15, Siglec-4 (MAG; myelin-associated glycoprotein) also preferentially binds clustered *O*-glycan ligands ST and dST.[294] Siglec-4 specifically requires α2−3 sialylation of core 1, and unlike other Siglecs it is expressed by oligodendrocytes and Schwann cells where it plays a role in maintenance in myelinated axons and regulation of neural outgrowth.[314] However, Siglec-4 has also proven to be involved in pancreatic cancer, where it promotes perineural invasion by enhancing adhesion between Schwann cells and glycosylated MUC1 expressed by the pancreatic cancer cells.[315] As such Siglec-4 is not an immune checkpoint, but an adhesion molecule whose glycan recognition properties allow tumors to spread.

Siglec-9 was initially proposed to preferentially bind to ST *O*-glycans,[316] but has more recently been shown to bind all terminally α2−3 sialylated structures including α2−3 sialylated LacNAc present on many different types of glycoconjugates.[294,309] A likely role of Siglec-7 and Siglec-9 in cancer progression is supported by studies showing that increased expression α2−3 sialylated structures on pancreatic cancer cells promotes differentiation of immunosuppressive TAMs.[309,316,317] Less is known about Siglec-10, although it is known to interact with the heavily *O*-glycosylated protein CD24 upregulated in multiple tumors and function in protection against immune mediated tissue damage and mediate immune tolerance in the placenta.[318,319] Siglec-10 is expressed by macrophages and reported to recognize α2−3 as well as α2−6 sialylated ligands.[320] Disruption of the interaction between Siglec-10 and CD24 reduces tumor growth in murine models through macrophage-dependent mechanisms,[321] and Siglec-10 expressing NK cells in hepatocellular carcinomas exhibit reduced effector functions and correlate negatively with patient survival.[322] Finally, the closely related Siglec pair, Siglec-5 and 14, have been shown to bind STn.[323] These Siglecs possess nearly identical ligand binding domains and are expressed largely in the same cell types, including neutrophils, monocytes, and B cells, but differ in signaling potential, as Siglec-5 contains ITIMs while Siglec-14 signals through ITAMs-associated adaptor proteins.[323] Their paired expression therefore likely contributes to secure a balanced immune response. While the function of Siglec-14 in cancer is unexplored, the murine Siglec-5 ortholog Siglec-F has been demonstrated to promote neutrophil infiltration in murine lung cancer models.[324] Whether this is recapitulated in humans remains to be established.

Taken together, these findings demonstrate that Siglecs mediate immunosuppressive and tumor-promoting functions through recognition of *O*-glycan epitopes. Although their exact binding specificities remain incompletely resolved, studies using mucins and mucin-like domains to present clusters of specific *O*-glycan structures clearly demonstrate that Siglecs seem to recognize SAMPs within mucin tandem repeats.[294,325,326] Accordingly, the dense sialoglycan clusters generated by concurrent upregulation of sialylation and heavily glycosylated carrier proteins such as mucins are likely to represent a major mechanism by which cancer cells establish immunosuppression. Siglecs, acting in concert with clustered sialylated O-glycans on mucin domains, therefore emerge as important immune checkpoints in cancer, and a deeper structural understanding of their ligand recognition will be essential for future therapeutic targeting.

3.3. MGL Receptor Mediates Immunosuppression through Tn-Recognition

The MGL receptor is another GBP involved as an immune checkpoint.[327] MGL is an endocytotic C-type lectin receptor (CLR) expressed selectively by dendritic cells (DCs) and macrophages, and it is the only known CLR with a clear specificity for the Tn-antigen.[328−330] In addition to Tn, MGL can also recognize STn and other terminal GalNAc structures (such as LacdiNAc and asialo-GM2), and to a lesser extent galactose, although the functional importance of the more elongated glycan-ligands is unclear.[184,331−336] In general, MGL is upregulated in tolerogenic DCs and TAMs,[337−339] and engagement of MGL by Tn-containing ligands promotes a tolerogenic response.[329,334,340−343] It has also been shown to induce T_H2 responses upon recognition of certain bacteria and helminths.[344−348] Importantly, MGL mediated internalization and signaling are highly sensitive to the structural nature of its ligand. For example, MGL-mediated internalization in DCs of large Tn-glycosylated recombinant MUC1 glycoproteins, consisting of 16 tandem repeats, was arrested in endolysosomal/HLA-II compartments,[349] whereas a shorter Tn-glycosylated MUC1 peptide consisting of only 3 TRs was delivered to HLA-I and HLA-II compartments.[329] Subsequent studies showed that stimulation of DCs with this shorter glycopeptide induced DC maturation and enabled IFN-γ production by CD8+ T cells, despite a cytokine profile skewed toward a tolerogenic IL-12/IL-10 balance.[340] Similarly, a longer Tn-glycosylated 100-mer MUC1 peptide activated immature DCs, though the DCs did not induce a T_H1 response when cocultured with $CD4^+$ T cells.[350] Together, these findings show that, much like Siglecs, MGL function is shaped by ligand affinity, avidity, and structural context. Consistent with this, proteins carrying densely clustered Tn epitopes are readily taken up by antigen-presenting cells and can induce tolerogenic responses in murine models, most likely through MGL.[341] In cancer, MGL is present on APCs in the tumor stroma where its interactions with tumor-associated glycans supports the immunosuppressive tumor microenvironment. Activation of MGL receptor in DCs and macrophages induces secretion of the anti-inflammatory IL-10, resulting in inhibition of $CD4^+$ T cell proliferation and induction of suppressive IL-10 secreting $CD4^+$ T cells.[328,351,352] Furthermore, MGL expressing APCs recognize CD45 on effector T cells, modulate its phosphatase activity, and interfere with TCR signaling, thereby reducing T-cell proliferation and promoting CD45-dependent cell death.[353] While this finding may indicate the presence of Tn-antigen on CD45 in activated T cells, the molecular basis for this observation remains unresolved. In colon carcinoma, MGL expressed by stromal APCs has been shown to bind MUC1,[338] and the presence of MGL expressing cells in stage III colorectal, cervical squamous cell, and adeno-squamous carcinomas are correlated with poor disease-free survival. Furthermore, expression of MGL ligands (i.e., Tn) has been associated with V600E mutated BRAF activity in stage III colorectal tumors and with PIK3CA mutations in cervical tumors.[339,354] Collectively, these findings support a direct role for Tn expression in cancer progression, not only as a marker of aberrant glycosylation, but also as a functional determinant of immune tolerance and immune checkpoint. By engaging MGL on stromal immune cells, Tn-bearing glycoproteins can help establish an immunosuppressive tumor microenvironment, which may in part explain the widespread expression of Tn-antigen across human cancers, and why this feature is often lost in cancer cell lines.

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3.4. Galectins Regulate Various Processes and Immune Homeostasis in Cancer

Galectins represent another class of GBPs with important roles in oncogenesis through their effects on angiogenesis, apoptosis, autophagy, proliferation, and immune regulation.[355−360] Upon ligand binding many galectins can form multivalent lattices at the cell surface that cluster glycoproteins and receptors and thereby regulate their retention, signaling, and endocytosis.[361] All galectins bind β-galactoside containing glycoconjugates, such as LacNAc and poly-LacNAc, although differences in domain organization and ligand specificity give rise to diverse biological functions.[362,363] Additionally, several galectins, particularly galectin-3 but also galectin-1, 2, 4, and 8 have been shown to recognize the T-antigen.[364] Galectins are synthesized in the cytoplasm and can function intracellularly, but they can also accumulate at the cell surface or in the extracellular matrix despite lacking classical signal peptides.[357] Their export is therefore noncanonical and incompletely understood, although exosome-dependent export has been demonstrated for Gal-3,[365] and Gal-1 has been reported to be released by secretory-autophagy from TAMs in hepatocellular carcinoma.[366] In addition, Gal-9 has been shown to be secreted by regulated exocytosis in AML.[367] In tumors, galectins are produced not only by cancer cells but also by endothelial cells, stromal cells and immune cells.[357,359,360]

Galectin-1 is among the best-characterized galectins in cancer, associated with poor survival, and consistently linked to suppression of antitumor immunity.[360,368,369] Galectin-1 regulates both innate and adaptive immune responses. This includes inhibition of trans-endothelial migration of effector lymphocytes through receptors such as CD43, and reinforcement of immunosuppressive functions mediated by regulatory T cells.[370−375] Furthermore, galectin-1 has been shown to suppress Th1-type responses in favor of Th2- and Treg-associated activation.[370−375] Galectin-1 has also been implicated in regulation of T-cell apoptosis by binding preferentially to core 2 O-glycosylated CD45[376−378] and by sensitizing activated T cells to Fas/FasL- and caspase-dependent death pathways.[379,380] In mice, interactions with core 2 structures on T-cells attenuated CD8 T-cell responses and protected from colitis.[381] Additionally, galectin-1 directly promotes tolerogenic phenotypes of APCs[382] by reducing their antigen-presenting capabilities,[383] and through induction of suppressive cytokines, IL-10- and IL-27, in monocytes and dendritic cells.[384,385] In macrophages, galectin-1 stimulates the expression of IDO and PD-L1 and promotes a broader M2/TAM-like program characterized by increased arginase-1, IDO1, and PD-L1.[386] Recent work further shows that galectin-1 can coordinate both immunoregulatory and pro-angiogenic programs in myeloid-derived suppressor cells, reinforcing its role as a regulator of the tumor microenvironment.[387] Importantly, macrophages themselves can also serve as a major stromal source of galectin-1 and in hepatocellular carcinoma, TAMs actively secrete galectin-1 through TLR2-dependent secretory autophagy.[366]

Galectin-3 likewise plays pleiotropic and highly context-dependent roles in cancer.[360] Galectin-3 can oligomerize to form large lattice structures, and its binding and function may therefore be highly dependent on ligand density.[388−390] In addition to the classic LacNAc recognition, galectin-3 also binds MUC1 and MUC16 in a galactose-dependent manner, possibly through clustered T antigen or related core 1 O-glycans.[391] Consistent with this, galectin-3 has been shown to bind MUC1 glycosylated with T antigen to promote endothelial adhesion of multiple cancer cell lines,[364,392,393] as well as enhance EGFR activation through MUC1-dependent interactions.[394] In melanoma, galectin-3 binds MUC18 and promotes its clustering at the cell surface, thereby inducing Akt signaling and enhancing proliferation, migration, adhesion, and invasion.[395] Galectin-3 secretion is reported to be induced in tumor-associated macrophages (TAMs) by hypoxia, and inhibition of galectin-3 inhibited breast cancer progression in murine models.[396] Although downregulation of galectin-3 has been reported in some tumors, including prostate and basal cell carcinomas,[360] such findings might reflect proteolytic cleavage of galectin-3,[397] or redistribution between nuclear, cytoplasmic, and extracellular pools. Tumors often show increased cytoplasmic expression of galectin-3,[398,399] which could be explained by increased lysosomal stress in cancer cells due to the function of galectin-3 in lysosomal repair.[400] Importantly for cancer, galectin-3 has a dual effect on immunity: it promotes innate immune activation through recruitment of granulocytes, monocytes, and macrophages[401−404] while simultaneously suppressing adaptive immunity by inducing T-cell apoptosis and inhibiting T-cell receptor signaling through formation of multivalent complexes that interfere with TCR assembly.[405,406] More recent work further suggests that galectin-3 can suppress activated CD8 T cells and reprogram monocyte/macrophage populations toward a suppressive phenotype, for example through TREM2-associated pathways in lung cancer.[396,407] This combination of proinflammatory and immunosuppressive activities, together with its compartment dependence, may allow tumors to exploit galectin-3 to establish an inflammatory microenvironment while simultaneously evading adaptive immune responses.

Galectin-8 is less comprehensively defined in cancer than galectin-1 and galectin-3, but emerging evidence suggests important tumor-intrinsic and myeloid-regulatory functions.[408] Increased galectin-8 has been reported in breast, prostate, lung, and colon cancer.[409] Functionally, extracellular galectin-8 can promote adhesion of tumor cells to endothelium and induce cytokine- and chemokine-signaling that supports formation of metastatic niches.[409] At the same time, its role appears highly context dependent, as galectin-8 has also been linked to reduced migration and EMT in colorectal cancer[410] and to improved outcome in some breast cancer settings.[411] Another example is the function of galectin-8 in immune regulation. Through binding of galectin-8 to the immune checkpoint receptor leukocyte immunoglobulin-like receptor B4, present on myeloid cells in solid tumors, galectin-8 activates STAT3 while inhibiting NF-κB, converting the myeloid cells into potent suppressors of the immune response.[408]

Finally, galectin-9 is also upregulated in cancer and produced by both tumor cells and immune cells.[412] Extracellular galectin-9 suppresses Th1/Th17 and cytotoxic CD8+ T-cell responses and promotes Treg differentiation, classically through the heavily O-glycosylated immune checkpoint TIM-3.[413−418] In addition to its actions on lymphocytes, galectin-9 has important myeloid effects. In pancreatic ductal adenocarcinoma, galectin-9 binds Dectin-1 on macrophages and promotes tolerogenic macrophage programming and adaptive immune suppression.[419] Recent work further suggests that galectin-9 can support IL-10-producing immunosuppressive macrophage states across multiple cancer types.[420] The clinical effect of galectin-9 expression, however, again seems to be highly contextual. In AML and glioblastoma, higher galectin-9 is generally associated with tumor progression and poor outcome, whereas in

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Table 2. Clinical Trials Targeting O-Glycoproteins or O-Glycan TACAs

Target	Name	Type	Condition	Phase/Status	Trial ID
Glycoproteins					
CEA	ETBX-011	Adenoviral vaccine	Colon, lung, and breast cancer	Phase I/II, completed 2013	NCT01147965
CEACAM5/6	NEO-201	Monoclonal antibody	NSCLC, head and neck squamous cell carcinoma, cervical cancer, uterine cancer	Phase I/II, recruiting	NCT03476681
Mesothelin	Mesothelin CAR T cells	CAR T cells	Lung cancer	Phase I, ongoing	NCT03198052
MUC1	huMNC2-CAR44 CAR T cells	CAR T cells	Breast cancer	Phase I, ongoing	NCT04020575
	Anti-MUC1 CAR T cells	CAR T cells	NSCLC	Phase I/II, unknown 2022	NCT03525782
	MUC1 CAR T cells	CAR T cells	Lung cancer	Phase I, ongoing	NCT03198052
			Intrahepatic cholangiocarcinoma	Phase I/II, ongoing	NCT03633773
	MUC1 Peptide-Poly-ICLC	Peptide vaccine	Lung carcinoma	Phase I, ongoing	NCT03300817
			Colorectal adenoma	Phase II, ongoing	NCT02134925
	Mannan-MUC1-pulsed DC	Dendritic cell vaccine	Adenocarcinoma	Phase I, completed 2006	PMID 16467101
	Oxidised mannan-MUC1	Vaccine	Breast cancer	Pilot phase III, completed 2013	ISRCTN71711835
MUC16	Abagovomab	Monoclonal antibody	Ovarian cancer	Phase II/III, terminated 2011	NCT00418574
	Oregovomab	Monoclonal antibody	Ovarian neoplasm	Phase II, completed 2018	NCT01616303
			Ovarian, fallopian tube, and peritoneal cancer	Phase III, ongoing	NCT04498117
			Ovarian cancer	Phase III, terminated 2007	NCT00050375
PSCA	PSCA CAR T cells	CAR T cells	Lung cancer	Phase I, ongoing	NCT03198052
Truncated Glycan Structures					
Tn-MUC1	CART-TnMUC1 (5E5)	CAR T cells	Multiple myeloma, NSCLC, ovarian, fallopian tube, pancreatic duct, and triple negative breast cancer	Phase I, terminated 2024	NCT04025216
TA-MUC1	PankoMab-GEX	Monoclonal antibody	Solid tumors	Phase I, completed 2013	NCT01222624
			Ovarian, fallopian tube, and peritoneal cancer	Phase II, completed 2017	NCT01899599
			Solid tumors	Phase I, completed 2020	NCT03360734
	DS-3939a	Antibody-drug conjugate	Solid tumors	Phase I/II, recruiting	NCT05875168
Tn	Tn(c)-KLH	Carbohydrate vaccine	Prostate cancer	Phase I, completed 2016	PMID 14645418
	MAG-Tn3	Carbohydrate vaccine	Breast cancer	Phase I, completed 2021	NCT02364492
STn	Theratope	Carbohydrate vaccine	Breast cancer	Phase II, completed 2005	NCT00046371
			Breast cancer	Phase III, completed 2008	NCT00003638
	sTn(c)-KLH	Carbohydrate vaccine	Breast cancer	Phase I, completed 2007	PMID 10353726
	huCC49	Radiolabeled antibody	Ovarian and prostate cancer	Phase I/II, completed 1996−2001	PMIDs 8790200, 9103398, 10541372

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Table 2. continued

Target	Name	Type	Condition	Phase/Status	Trial ID
Truncated Glycan Structures					
	CART72	CAR T cells	Colorectal cancer	Phase I, completed 2017	C-9702
	TAG72 CAR T cells	CAR T cells	Ovarian cancer	Phase I, ongoing	NCT05225363
	B72.3	Radiolabeled antibody	Ovarian cancer	Phase I, completed 1999	PMID 10353726
T	TF(c)-KLH QS21	Carbohydrate vaccine	Prostate cancer	Phase I, completed 2009	NCT00003819
	Y90 MOAB m170	Radiolabeled antibody	Breast cancer	Phase I, unknown 2013	NCT00009763
			Prostate cancer	Phase I, unknown 2013	NCT00009750
Polyvalent	Globo H−GM2−sTn−TF−Tn−KLH	Carbohydrate vaccine	Fallopian tube, ovarian, and peritoneal cancer	Phase I, completed 2017	NCT01248273
	P10s (Le^y, GD2, GD3, GM2)	Carbohydrate mimetic vaccine	Breast cancer	Phase I, completed 2019	NCT01390064
	Globo-H-GM2-Le^y-MUC1−32-sTn-TF-Tn-KLH	Carbohydrate vaccine	Breast cancer	Phase NA, completed 2009	NCT00030823
	ETBX-011 (CEA), ETBX-061 (MUC1), ETBX-051 (brachyury)	Combined adenoviral vaccine	Colon, lung, breast, and prostate cancer	Phase I, completed 2020	NCT03384316
Complex Glycan Structures					
CanAg	huC242	Antibody-drug conjugate	Noncolorectal, pancreatic cancer	Phase I, completed 2009	NCT00352131
CA6	SAR566658	Antibody-drug conjugate	Malignant neoplasms	Phase I, completed 2017	NCT01156870
			Triple negative breast cancer	Phase II, terminated 2018	NCT02984683
Fucosylation	SGN-2FF	Small molecule (antagonist)	NSCLC, renal cell, head and neck, gastric, and gastro-esophageal junction carcinomas, breast, urinary bladder, and colorectal neoplasms	Phase I, terminated 2019	NCT02952989
Sialylation	E-602	Sialidase therapy	Melanoma, ovarian, NSCLC, colorectal and pancreatic cancer	Phase I/II, completed	NCT05259696
SLe^a	MVT-5873	Monoclonal antibody	Pancreatic cancer	Phase I, terminated	NCT02672917
			Colon and pancreatic cancer, cholangiocarcinoma, metastatic colon cancer, liver metastasis	Phase II, completed 2023	NCT03801915
	MVT-1075	Radiolabeled antibody	Pancreatic cancer, SLe^a positive tumors	Phase I, terminated 2023	NCT03118349
	MVT-2163	Radiolabeled antibody	Pancreatic cancer, SLe^a positive tumors	Phase I, terminated 2017	NCT02687230
	sLe^a-KLH	Carbohydrate vaccine	Breast cancer	Pilot study, completed 2020	NCT00470574
Le^y	Le^y-KLH	Carbohydrate vaccine	Ovarian cancer	Phase I, completed 2000	PMID 10861456
	BR96-doxorubicin	Antibody-drug conjugate	Epithelial tumors	Phase I, completed 2000	PMID 10829049
			Breast cancer	Phase II, completed 1999	PMID 10080588
	SGN-10	Antibody-drug conjugate	Solid tumors	Phase I, completed 2002	PMID 12374676
	SGN-15	Monoclonal antibody	NSCLC	Phase II, completed 2003	NCT00051571
	LeY T cells	CAR T cells	Acute myeloid leukemia	Phase I, completed 2013	CTX 08−0002
	Le^y CAR T cells	CAR T cells	Lung cancer	Phase I, ongoing	NCT03198052
	LeY-CAR-T	CAR T cells	Le^y expressing solid tumors	Phase I, completed 2022	NCT03851146
	Hu3S193	Monoclonal antibody	SCLC	Phase I, completed 2006	NCT00084799

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Table 2. continued

Target	Name	Type	Condition	Phase/Status	Trial ID
Complex Glycan Structures					
			Fallopian tube, ovarian, and peritoneal cancer	Phase II, completed 2012	NCT00617773
Le^y, Le^b	GNX102	Monoclonal antibody	Solid tumors	Phase I, terminated 2023	NCT04250597
Glycan-Binding Proteins					
E-selectin	Uproleselan	Small molecule (antagonist)	Multiple myeloma	Phase I, completed 2019	NCT02811822
			Acute myeloid leukemia	Phase I/II, completed 2018	NCT02306291
				Phase II/III, suspended 2024	NCT03701308
				Phase III, terminated	NCT03616470
				Phase I, ongoing	NCT04964505
	GMI-1359	Small molecule (antagonist)	Breast cancer	Phase I, terminated 2022	NCT04197999
Galectin 1,3	GR-MD-02	Galectin antagonist	Metastatic melanoma	Phase I, completed 2018	NCT02117362
			Melanoma, NSCLC, head and neck squamous cell carcinoma	Phase I, completed 2023	NCT02575404
	GM-CT-01	Galectin antagonist	Metastatic colorectal cancer	Phase II, completed 2008	NCT00110721
			Colorectal, lung, breast, head and neck, and prostate cancer	Phase I, completed 2006	NCT00054977
			Metastatic melanoma	Phase I/II, terminated 2019	NCT01723813
Galectin 3	GB1211	Galectin antagonist	NSCLC	Phase I/II, ongoing	NCT05240131
Galectin 9	LYT-200	Galectin antibody	Solid tumors	Phase I/II, completed	NCT04666688
			AML, MDS	Phase I, recruiting	NCT05829226
Siglec-2	Epratuzumab	Monoclonal antibody	Diffuse large B cell lymphoma	Phase II, completed 2015	NCT00301821
			Follicular lymphoma	Phase II, completed 2014	NCT00553501
			Acute lymphoblastic leukemia	Phase II, completed 2017	NCT00945815
	Pinatuzumab vedotin	Antibody-drug conjugate	Follicular lymphoma and diffuse large B cell lymphoma	Phase I/II, completed 2019	NCT01691898
	Inotuzumab ozogamicin	Antibody-drug conjugate	Acute lymphoblastic leukemia	Phase III, completed 2017	NCT01564784
	CD19/CD22 CAR T cells	CAR T cells	Acute lymphoblastic leukemia	Phase I, ongoing	NCT03233854
	CD22 CAR T cells	CAR T cells	Hairy cell leukemia	Phase I, ongoing	NCT04815356
			B cell malignancies	Phase I, ongoing	NCT04088890
Siglec-3	Gemtuzumab ozogamicin	Antibody-drug conjugate	Acute myeloid leukemia	Phase II, completed 1999	NCT00003673
				Phase II, completed 2002	NCT00003131
				Phase III, completed 2013	NCT00927498
				Phase III, completed 2014	NCT00085709
Siglec-15	NC318	Monoclonal antibody (antagonist)	Metastatic solid tumors, head and neck squamous cell carcinoma, NSCLC, ovarian cancer, triple-negative breast cancer	Phase I/II, completed 2024	NCT03665285

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Table 2. continued

Target	Name	Type	Condition	Phase/Status	Trial ID
Glycan-Binding Proteins					
			Advanced NSCLC	Phase II, ongoing	NCT04699123

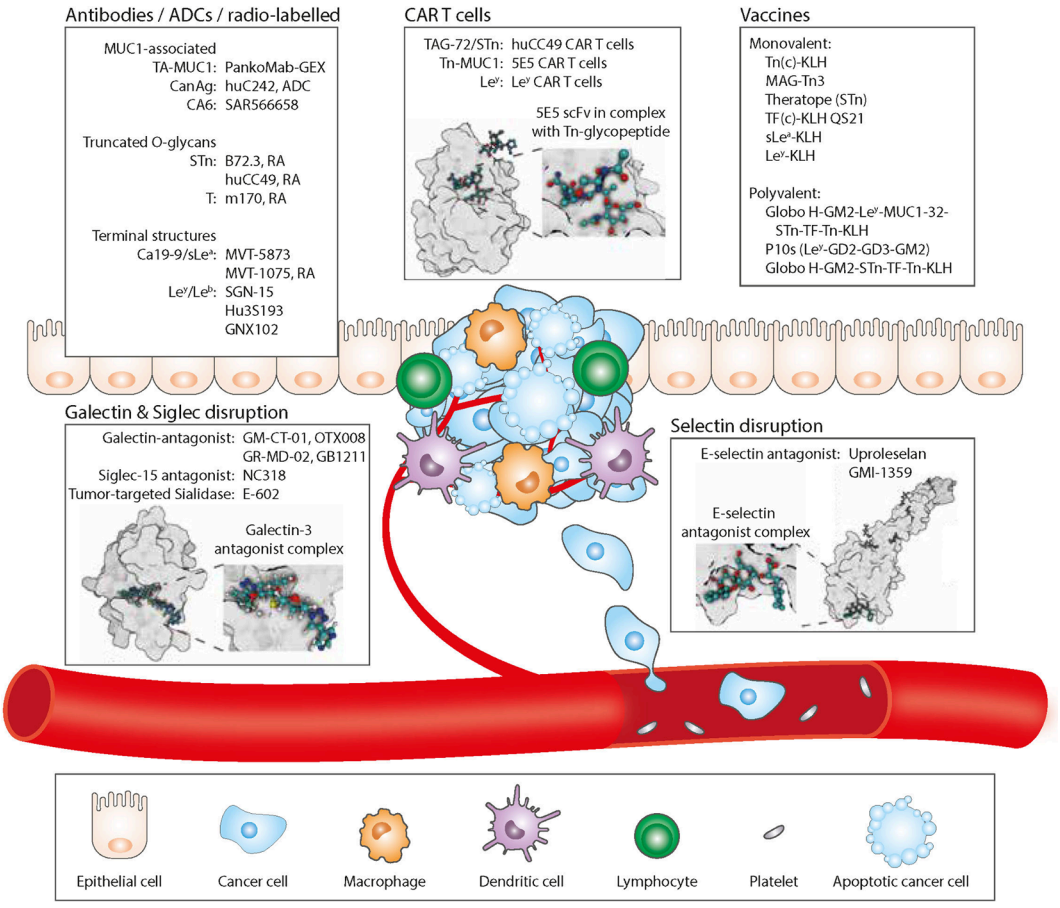


Figure 4. This figure illustrates how several different strategies have been developed to target or disrupt *O*-GalNAc glycosylation in cancer therapies. Cancer-associated glycoproteins, mainly MUC1, and TACAs, such as truncated Tn, STn, and T and terminal sLe^y, have been directly targeted using several different strategies. In attempts to vaccinate patients against these structures, monovalent and polyvalent vaccines have been generated, presenting either a single structure or several different at once. Several monoclonal antibodies have been developed against Tn-MUC1, STn, T, and Lewis antigens, which have been tested in clinical trials either unconjugated, as antibody-drug conjugates (ADCs), or radiolabeled antibodies (RAs). Furthermore, some antibodies have been adapted to scFv format to generate CAR T cells (example 3D structure: 5E5 scFv, PDB 6TNP). Direct targeting of cancer cells expressing Siglec-2 and 3 has also undergone clinical testing, and endeavors to disrupt the immunosuppression elicited by Siglecs and sialoglycans has led to the generation of a Siglec-15 antagonistic antibody and a novel sialidase therapy. Immunosuppression elicited by galectin interactions have likewise been targeted by galectin antagonists (example 3D structure: Galectin-3 antagonist complex, PDB 7DF5), while attempts to disrupt selectin-mediated metastasis has resulted in the development of an E-selectin antagonist (example 3D structure: E-selectin antagonist complex, PDB 4C16).

melanoma loss of galectin-9 has been linked to progression and metastasis.[412,421]

Taken together, these findings suggest that, while galectin-1 is the most tumor-promoting galectin in the immune microenvironment and has clear immune checkpoint-inhibitory functions, galectin-3, galectin-8 and galectin-9 also show profound influences on cancer progression across tumor types and receptor systems.

3.5. Context and Avidity Shape How Cancer-Associated *O*-GalNAc Glycans Engage Glycan-Binding Proteins

As an overall conclusion, a major way in which cancer-associated O-GalNAc glycans promote tumor progression is through their interactions with glycan-binding proteins. Moreover, it is evident that glycan ligands cannot be viewed as freely accessible motifs defined solely by glycan composition. Rather, they must be understood in context: as features presented on a specific protein carrier, at a defined site, and within a protein scaffold whose conformation may itself change upon receptor activation.[422,423] As such, the functional ligands of several glycan-binding proteins may be rapidly altered or unmasked by neuraminidase-dependent desialylation or by conformational rearrangements.[390,422,424] For example, recent work shows that galectin-3 can respond to the spatial presentation of glycans on a specific protein scaffold, providing a conceptual model for context-dependent glycan recognition.[424] A central challenge for the field is therefore not only to determine which glycans are present on a cell or a specific protein, but also which of them are exposed and recognizable within the structural context of the

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tumor cell surface. A second element that deserves greater attention is avidity. The function of glycan-binding proteins can change dramatically when the same glycan motifs are displayed at high density, for example on extended mucin domains.[316,425] Under such conditions, glycan-binding proteins that normally interact only weakly with a given glycan may bind efficiently when that glycan is presented in dense clusters, thereby generating high-avidity interactions. In this way, arrays of individually low-affinity ligands may become functionally dominant. This is seen for Selectins, for Siglecs, for the MGL receptor, and may be particularly important for oligomerizing lectins such as galectin-3, which can form lattices and prolong receptor residence at the cell surface.[406,426] Upregulated mucins or mucin domains in receptor proteins may therefore act as ligands that redirect receptor clustering, in some cases promoting activation and endocytosis, and in other cases inhibiting activation or even function as molecular sinks that sequester lectins away from membrane-proximal receptors or immune synapses.[308,427] Future progress will therefore depend on integrating glycoproteomics with spatially and kinetically resolved measurements on intact cell surfaces in order to define how glycan density, accessibility, and carrier context reconfigure glycan-binding protein biology in cancer. Some emerging methods may soon enable this type of analysis. For example, the scGOAT-seq and GlycoScope methods using human lectins to integrate functional glycan accessibility with single-cell and spatial multiomic measurements as well as other spatial approaches demonstrate that the field is definitely moving in this direction.[44,428]

4. THERAPEUTIC STRATEGIES TARGETING ABERRANT *O*-GALNAC GLYCOSYLATION IN CANCER

Targeting *O*-glycan TACAs offers an attractive strategy for accessing solid tumors, as these antigens are abundant and often exclusively expressed on the surface of cancer cells. Furthermore, blocking interactions between *O*-glycans and glycan-binding proteins may help prevent metastasis or serve as a novel class of checkpoint inhibitors that enhance immune activation. Over the past decades, numerous *O*-glycan-targeting therapies have been developed and evaluated in clinical trials, including vaccines, monoclonal antibodies (mAbs), and their derivatives, such as antibody-drug conjugates (ADCs), chimeric antigen receptor (CAR) T cells, CD3-redirecting T-cell bispecific antibodies (TCBs), radiopharmaceuticals, and glycocalyx and extracellular matrix-remodeling enzymes. In addition, small-molecule inhibitors have been developed targeting glycan-binding proteins. Although early *O*-glycan-targeting strategies failed to meet major clinical end points, new approaches are now being developed, and promising clinical trials are underway with this next generation of more potent immunotherapeutic agents (Table 2, Figure 4).

4.1. *O*-GalNAc Glycan Vaccination Strategies

Much research originally focused on generating therapeutic cancer vaccines using truncated *O*-glycans, targeting mainly the Tn-, STn-, and T-antigens. Initial strategies utilized rather crude preparations as immunogens, resulting in unsatisfactory outcomes. The first *O*-glycan vaccine was developed by Springer and colleagues following their identification of Tn- and T-antigen as carcinoma markers and was based on membrane extracts from red blood cells. The results demonstrated a potential effect on disease free survival but were ultimately deemed inconclusive due to lack of control patients.[429−431] Similarly, a vaccine derived from desialylated ovine submaxillary gland mucin (OSM) was found to only induce low affinity IgM and low IgG titers, though it was well-tolerated.[432] To overcome the low immunogenicity and need for more standardized vaccines, synthetic monovalent vaccines were developed by conjugating Tn, STn, or T-antigen to a molecular carrier. Clinical studies testing T-KLH and STn-KLH vaccines yielded varying titers and affinities of antibodies to the immunogen in ovarian and colorectal cancer patients, but failed to induce potent reaction with the tumor cells.[433,434] An alternative STn-KLH vaccine was developed, Theratope, achieving high valency by conjugating several STn-antigens to the KLH carrier. Theratope achieved encouraging results in the initial phase I and II trials, including clinically relevant antibody titers and increased survival of patients with and without cyclophosphamide treatment.[433,435−441] However, in the phase III trial which enrolled more than 1000 metastatic breast cancer patients, Theratope failed to meet the clinical end points despite producing good antibody titers in the patients.[442] It is worth mentioning that patients were not prescreened for STn expression in their tumors, which might be an explanation for the failure of this study.[443] Furthermore, antibody titers were shown to be significantly higher in patients receiving pretreatment with cyclophosphamide,[439] an agent known to exhibit significant immunomodulatory effects through selective depletion of regulatory T cells. It could therefore be interesting to revisit Theratope in trials with improved patient screening and in combination with novel immune checkpoint inhibitors. Despite the clinical failure of Theratope, it served as an important proof-of-concept that glycan vaccines require proper valency for induction of strong immune responses. To develop vaccines with an improved valency and presentation of TACAs, several multivalent vaccines were developed where multiple STn-antigens mimicking clustered *O*-glycans were conjugated to KLH,[444] inducing significant IgM and IgG titers, although only the generated IgM antibodies were able to stain cancer cells sufficiently.[445] Multivalent vaccines targeting Tn- and T-antigen were similarly developed, where trimeric Tn or T was conjugated to either KLH or palmitic acid (PAM).[446,447] In a phase I clinical trial in patients with relapsed prostate cancer, these vaccines appeared to have antitumor effects as seen by a decrease in the prostate-specific antigen (PSA) marker even though the developed IgG antibodies seemed unable to bind to cancer cells.[446,448] Other vaccines were developed aimed to raise antibody responses to multiple TACAs simultaneously. One such vaccine consisted of Tn-MUC2 and Globo-H bivalently conjugated to KLH successfully induced both IgM and IgG titers against both glycan structures in relapsed prostate cancer patients.[449] Similarly, a heptavalent vaccine was developed with seven individual immunogens conjugated to KLH (GM2; Globo-H; Le^y; Tn-; STn-; T/Tn-MUC1), which in a clinical trial induced IgM titers against Globo-H, Tn, STn, T, and Tn-MUC1 immunogens, only Tn-MUC1 induced IgG titers.[450]

Ultimately, these vaccine strategies failed, likely because they were unable to induce high-affinity IgG responses. Indeed, as will be discussed shortly, it has become increasingly clear that high-affinity antibodies are more easily raised against glycopeptide epitopes rather than TACAs themselves. This is also consistent with the autoantibody profiles in human cancer patients evaluated with glyco-peptide arrays.[103,112] Several technologies including cell-based, chemoenzymatic, and chemical synthesis strategies have emerged as production platforms to supply glycopeptides for use in vaccine development.[451−454]

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This has allowed for increased control and optimization of glycopeptide epitopes and conjugation strategies. For example, a fully synthetic tripartite vaccine combining a Tn-modified MUC1 glycopeptide, a helper-T-cell epitope, and Pam3CysSK4 elicited tumor-reactive anti-MUC1 IgG responses.[455] These antibodies were driven by the aberrantly glycosylated MUC1 epitope and were consistent with recognition of the immunodominant PDTR-containing region. Other examples include the use of glycopeptide arrays helping to define the optimal MUC1 glycopeptide antigens used to generate improved IgG titers with tumor protective effects in mice.[456,457] In these studies, the glycopeptides were presented on virus-like particles, which highlights the possibility to use nanoparticles including liposomes, virus-like particles, and gold nanoparticles, for improved epitope presentation.[458−460] The Westerlind group found a 10-fold increase in potency by conjugating synthetic MUC1 glycopeptides to PEGylated gold nanoparticles,[461] while the Barchi group successfully directed a vaccine to APCs for increased immune response by conjugating a synthetic MUC4 glycopeptide to gold nanoparticles coated with β-glucans.[462] Collectively, all these studies demonstrate how vaccination strategies have evolved from utilizing relatively crude cell- and protein products, to using highly defined, synthetic glycopeptides. Finally, dendritic cell vaccines targeting MUC1 have also shown promise. In a phase I trial, autologous DCs loaded ex vivo with recombinant MUC1 fused to highly mannosylated and fucosylated yeast mannan elicited a strong TH1 response in 9 of 10 patients with MUC1-positive adenocarcinomas.[463] In a subsequent randomized, double-blind study in 31 breast cancer patients, no recurrences were observed in vaccinated patients after 5.5 years, and after 15 years recurrence was seen in only 13% of vaccinated patients compared with 60% in the placebo group.[464,465] Although these studies used nonglycosylated MUC1, the same strategy could be adapted to glycosylated mucins carrying defined O-glycan epitopes.

In conclusion, although no clinical studies have been able to successfully employ TACA vaccines as a therapeutic strategy against cancer, much has been learned that may lead to the development of more successful vaccines, especially in combination with novel checkpoint inhibitor therapies, in the coming years. It is clear that glycan valency and conformational context are key for proper immunization, aspects that the use of synthetic glycopeptides displayed on multivalent nanoparticles, which could enable the development of effective curative, or perhaps more realistically, preventive vaccines targeting cancer-associated glycans.

4.2. Development of Monoclonal Antibodies against Tn, STn, and Lewis Antigens

As with the different vaccine strategies, considerable effort has been devoted to the development of monoclonal antibodies targeting the Tn, sTn, and Lewis antigens for both diagnostic and therapeutic applications (Supporting Information, Table 2). Many of these antibodies showed early promise and were initially thought capable of mediating antibody-dependent cellular cytotoxicity (ADCC). However, as with the carbohydrate vaccines, it soon became evident that naked antibodies alone were insufficient to mediate robust therapeutic efficacy, likely due to factors such as low antibody affinity, insufficient and variable antigen presentation, and tumor heterogeneity. As a result, subsequent efforts focused on functionalized antibody-based strategies to enhance tumor cell killing. Importantly, unlike vaccination approaches, mAbs and other binders to cancer-associated O-glycans can also serve as highly specific delivery platforms for potent therapeutic modalities, including ADCs, TCBs, CAR-T therapies, radiopharmaceuticals, or glycocalyx and extracellular matrix-remodeling enzymes, as mentioned above. In the following sections, we review the development of antibodies against Tn, sTn, and Lewis antigens, with particular emphasis on their therapeutic applications.

4.3. Direct Targeting of Tn- and Tn-Glycopeptides with Monoclonal Antibodies

Although numerous anti-Tn antibodies have been generated (Supporting Information, Table 2), such as CU-1,[466] 1E3,[467] 5F4,[468] MLS128,[469] BRIC66 and BRIC111,[470] 2154F12A4,[471] most remain largely diagnostic or epitope-mapping reagents and not advanced into a dedicated evaluation of their potential therapeutic potential. A smaller subset of Tn-directed mAbs, all generated using crude immunization strategies, were investigated more directly. Among these, the Tn-specific mAb GOD3−2C4 was raised with a combinatorial regime consisting of both asialo-OSM and the human lung cancer cell line A549.[472] It bound to multiple human carcinomas and was able to induce ADCC both *in vitro* and *in vivo*.[472] Likewise, the mAb 83D4 was generated from immunization with extracts from poorly differentiated human breast adenocarcinomas, and though it was shown to have a $K_D \sim 0.6$ nM toward asialo-OSM and a $K_D \sim 4$ nM toward synthetic glycopeptides carrying 2 or 3 Tn-antigens, it showed no reactivity against synthetic glycopeptide carrying a single Tn-antigen. This demonstrates the concept, that has become apparent in the later years, that some Tn antibodies bind to clusters of minimal of two neighboring GalNAc-residues.[389,473] To support clinical development, 83D4 was converted to a chimeric human/mouse IgG mAb, called Chi-Tn with similar specificity but slightly lower affinity ($K_D \sim 20$ nM toward synthetic glycopeptides carrying 2 or 3 Tn-antigens and $K_D \sim 9$ nM toward asialo-OSM).[473] Chi-Tn has been shown to bind selectively to various carcinomas, and though it showed no direct cytotoxicity *in vitro*, it was able to induce *in vivo* killing in an immunocompetent murine model in a macrophage and neutrophil dependent manner.[474] Furthermore, Chi-Tn has shown promising results *in vitro* and *in vivo* as an ADC conjugated to auristatin F, demonstrating rapid internalization and accumulation in solid tumors in murine models.[475] Another Tn antibody, KM3413 was generated through murine immunization of mucins purified from the human colon cancer cell line LS180, and similar to 83D4 and Chi-Tn it was found to preferentially bind clustered Tn-antigen (K_D = 160 nM) while not recognizing monovalent Tn-antigen.[476] Adapted to a human/mouse chimeric antibody, cKM3413 induced ADCC and improved survival in a murine xenograft model using highly Tn expressing Jurkat cells.[477] To make mAbs that only react with the monovalent Tn-antigen, a novel immunogen, the Tn-polysaccharide A1 (Tn-PS A1), was used. PS A1 is a MHC class II-binding zwitterionic polysaccharide, previously been used to induce T_H1/T_H17 responses.[478,479] With this strategy the IgM antibody, Kt-IgM-8, was generated specifically recognizing monovalent Tn-antigen independently of the peptide backbone. Kt-IgM-8induced CDC in MCF-7 cells *in vitro* and later *in vivo* in murine xenograft models.[480]

As an alternative to immunization with crude cellular lysates or isolated glycan structures, glycopeptide antigens can be used to generate antibodies directed against composite glycopeptide epitopes rather than glycans alone similar to what was described

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above for active vaccines. Such antibodies generally display higher affinity and specificity, likely because the peptide backbone constrains glycan orientation and thereby creates a more defined antigenic epitope.[103,112,481−484] This principle was first demonstrated using synthetic glycopeptides glycosylated with recombinant GalNAc-transferases, which gave rise to antibodies with high affinity and enhanced specificity.[485,486] As mentioned, this was later extended to the identification of naturally occurring autoantibodies against mucin O-glycopeptide epitopes in cancer patients,[103,112,113,487] used not only as a potential biomarker system, but also as a gateway for identifying biologically relevant glycopeptide antigens for the generation of novel Tn glycopeptide antibodies.[112] Along those lines, most initial attempts to generate glycopeptide-targeting antibodies were inspired by existing antibodies to mucin antigens, including MUC1. Several MUC1 mAbs, including HMFG1, HMFG2, SM3, C595, and AR20.5,[488−492] had shown to be influenced by the degree of MUC1 glycosylation, although epitope recognition was primarily determined by the peptide backbone. In particular, aberrant *O*-glycosylation of the threonine in the central MUC1 PDTR repeat created a conformational epitope recognized by several of the MUC1 antibodies.[491,492] For example, X-ray crystallography of AR20.5 showed that the Tn-antigen contributes little to the direct contact, but instead promotes a peptide conformation that enhances antibody binding.[493] This proved that cancer-associated *O*-glycans shape antibody recognition of the peptide backbone, but also demonstrated that the many MUC1 antibodies were not actual glyco-peptide specific, and explained why they maintained strong binding to healthy tissue.

Among the first examples of a genuinely glyco-peptide MUC1 specific mAb is PankoMAb.[494] PankoMAb was generated to recognize a tumor-associated conformational epitope within the central PDTR region of the MUC1 tandem repeat. Following immunization with aberrantly glycosylated MUC1, hybridomas were selected for reactivity against a glycosylated MUC1 peptide containing the PDT(-Tn)RP motif. Earlier descriptions often framed PankoMAb as a T/Tn-specific antibody to the PDTR region.[491,494] However, this appears too narrow. PankoMAb, and the derived optimized and humanized[495] antibody gatipotuzumAb (PankoMAb-GEX),[496] are more accurately described as a glyco-MUC1 antibody, and in line with this, the ADC, DS-3939a, formulated based on PankoMAb, shows strong binding to MUC1 carrying Tn, sTn, T, and ST, with minimal binding to the nonglycosylated peptide.[497] The nonarmed mAb GatipotuzumAb was well tolerated in phase I studies, but failed to improve progression-free survival as maintenance therapy in recurrent ovarian cancer in phase II.[496] This prompted the development of the ADC derivative DS-3939a from Daiichi mentioned above, which couples the humanized PankoMAb to a DXd topoisomerase I inhibitor payload.[497] DS-3939a is now in clinical development and promising data has been generated in a phase II trial. Some adverse events were observed, which may be due to cross reactivity with glyco-MUC1 expressed on healthy tissue, including the lung epithelia.

In parallel with these developments, several truly cancer selective Tn-MUC1 glycopeptide antibodies were generated by direct immunization with larger chemoenzymatically produced glycopeptide immunogens.[485,486] For example, immunization with a 60-mer Tn-glycosylated MUC1 peptide yielded the monoclonal antibody 5E5, which mainly recognizes the GST(-Tn)AAP epitope within the MUC1 tandem repeat, distinct from the epitopes recognized by AR20.5 and PankoMAb.[482] 5E5 binds with high affinity (KD = 1.7 nM) to Tn-MUC1, but also recognizes other Tn-glycosylated proteins.[484,485] It shows strong reactivity with several carcinomas, and induces CDC and ADCC in vitro.[484] It does not react with the surface of healthy tissue.[485,486] Conversion of 5E5 into a single-chain variable fragment enabled its incorporation into a second-generation CAR T-cell format with strong preclinical activity.[498] The 5E5 scFv retained specificity for MUC1-Tn, with no reactivity toward nonglycosylated peptides. Accordingly, 5E5 CAR T cells showed selective cytolytic activity in vitro and in vivo.[498] A small phase I clinical trial has evaluated 5E5 CAR T cells in 6 patients with solid tumors and in multiple myeloma (NCT04025216).[499] Early clinical findings showed biological activity and an encouraging safety profile, but only limited antitumor activity, possibly explained with highly variable target expression.[499] Interestingly, 5E5 has also been used in alternative approach to direct CAR T cells to Tn-expressing cancer cells by coexpressing a nonsignaling glyco-bridge, based either on 5E5 scFv or HPA lectin, in mesothelin directed CAR T cells. These glyco-bridge CAR T cells showed increased cytotoxicity by enhanced avidity toward Tn-expressing cancer cells, and showed antitumor activity against pancreatic PDX models, but still relied on target mesothelin expression to elicit cytotoxicity.[500] Another glycopeptide-directed CAR strategy was developed from the murine antibody 237Ab. This antibody recognizes a spontaneous murine fibrosarcoma antigen that was subsequently identified as a Tn-glycosylated ERGT(-Tn)-KPPLEELS epitope within OTS8, the murine ortholog of podoplanin (PDPN).[481,501] Although 237Ab itself showed strong antigen specificity, CAR T cells derived from its scFv were more promiscuous,[502] emphasizing that antibody specificity may not be fully preserved upon conversion into CAR formats. Crystal structures of 5E5 and 237Ab in complex with Tn-glycopeptides have confirmed that recognition is centered on the Tn-linked GalNAc residue. In both antibodies, the glycan is engaged mainly by the heavy chain, with contacts focused on the GalNAc hydroxyl groups and N-acetyl moiety, whereas the peptide contributes more to secondary shaping of the binding interface.[501,503] Other glycopeptide-selective antibodies include the dysadherin-Tn antibody 6C5 using isolated surface proteins from SimpleCell with homogeneous Tn expression.[504] More broadly, increasing knowledge of site-specific glycosylation has enabled the generation of a series of highly cancer specific antibodies against several defined Tn-peptide epitopes. This concept has been further advanced by GO Therapeutics, generating cancer selective glyco-peptide mAbs to mucin targets, including Tn-MUC1,[505] and Tn-MUC4,[506] but also in proteins such as LAMP,[507] cMET,[508] Fibronectin, and CD44.[508] These antibodies are mainly used to develop cancer-specific ADCs with potent efficacy in vitro and in vivo CDx and PDx models,[509−512] but also other targeted therapeutic modalities. One such example, is the antibody 4C8, recognizing a Tn-glycopeptide within the cancer-associated CD44 isoform CD44v6, and shown to direct CAR T-cell-mediated cytotoxicity both *in vivo* and *in vitro* using 3D tissue models created with mixtures of cancer cells and healthy cells.[513] Importantly, targeting the Tn-glycopeptide form of CD44v6 overcame known cross-reactivity with CD44v6 expressed in normal human skin, highlighting the selectivity and safety advantage that can arise from targeting composite glycopeptide epitopes rather than the protein backbone alone.[513]

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4.4. Targeting of STn-Antigen

Over time, a broad panel of mAbs recognizing the STn-antigen has been generated (see Supporting Information, Table 2). Similarly to the Tn antibodies, most of these antibodies were initially made using crude biological immunogens, including submaxillary mucins, tumor-cell membrane fractions, whole tumor cells, or tumor-derived glycoprotein preparations, while others were produced with more chemically defined antigens. Despite this broad antibody repertoire, only a few STn mAbs have advanced beyond diagnostic use and explored for therapeutic applications in murine models, and in a few cases in human studies. Among the earliest and most influential of these was B72.3,[514] and illustrates both the promise and the limitations of the first generation of STn-targeting antibodies. B72.3 was generated by immunization of membrane fractions of mammary carcinomas, and recognizes the tumor-associated glycoprotein (TAG) 72 epitope, an epitope later identified to be defined as STn-antigen.[515] The antibody displayed broad reactivity toward multiple carcinoma types, although reactivity to some normal tissues, particularly in the gastrointestinal tract, was also observed consistent with the known expression patterns of STn.[516,517] B72.3 subsequently entered into a phase I clinical trial as a radiolabeled chimeric antibody but development was limited due to anti-antibody responses in patients.[518,519] Despite limited therapeutic success, B72.3 was among the first FDA approved mAbs for whole body imaging of ovarian and colorectal patients as an indium-111 conjugate, Oncoscint (satumomAb pendetide),[520] although this product is no longer commercially available.

Given the early clinical interest in B72.3, the so-called second generation B72.3 antibodies were created by immunization with TAG-72 purified from colon cancer xenografts; one of which was called CC49.[521] CC49 was evaluated in phase I and II clinical trials as a ^{131}I-radiolabeled antibody in combination with interferon.[522] Although bone marrow suppression was observed as a dose-limiting toxicity and some patients initially experienced pain relief, no clear antitumor effects were observed.[523−526] Like the Tn-glycopeptide specific mAbs 5E5, 237, and 4C8, B72.3 and CC49 have both been adapted to CAR T cell format, originally as first-generation CAR constructs. B72.3 CAR T cells induced TAG-72 dependent IL-2 secretion and lysed TAG-72 positive metastatic colorectal adenocarcinoma cells in vitro.[527] Likewise, CC49 CAR T cells mediated specific lysis of TAG-72 expressing colorectal cancer cell lines and primary cancer cells *in vitro* and in murine *in vivo* studies.[528,529] However, a phase I clinical trial with CC49 CAR T cells was unsuccessful, likely due to lack of costimulation in the first-generation CAR construct, or due to immune responses against the murine CC49 scFv.[530] A second-generation CAR based on humanized CC49 and the 4−1BB costimulatory domain has since been developed and has shown to efficiently target ovarian cancer cell lines and primary cancer cells *in vitro* and *in vivo* in a murine ovarian solid tumor model.[531] Interestingly, recurrent tumors displayed signs of antigen escape marked by decreased TAG-72 expression.[531] A phase I trial of huCC49 CAR T cells in platinum resistant ovarian cancer patients has been initiated (Trial ID NCT05225363).

A challenge in the field is that the exact epitopes recognized by TAG-72/STn-reactive antibodies remain incompletely defined. Both B72.3 and CC49 appear to preferentially bind clustered STn-antigen, but their recognition is not limited to a single, structurally uniform epitope.[532] To increase a broader reactivity effort have been made to generate antibodies that recognize STn-antigen independently of peptide backbone using immunization with synthetic adjuvant-STn conjugates.[533] This approach led to the generation of multiple mAbs recognizing Neu5Ac in various configurations according to glycan array analysis, including 2G12−2B2 and 5G2−1B3 that both recognize STn-antigen.[533] These antibodies bound STn-expressing breast cancer cells and demonstrated *in vitro* efficacy and *in vivo* suppression of tumor growth in breast and colon murine xenograft models as MMAE-conjugated ADCs.[533] After humanisation, they induced complete tumor regression in CDX and PDX models of human ovarian cancer and showed a favorable toxicity profile in cynomolgus monkey toxicity studies.[534] However, no follow-up studies have yet been published, and no clinical trials have been registered.

Several additional STn mAbs are currently being developed guided by structural insight. Recent work on the anti-STn mAb L2A5, for example, provided high-resolution structural evidence that binding is mainly driven by the Neu5Ac residue of the core sTn epitope, with GalNAc adding stabilizing contacts, and suggests that tumor selectivity also depends on the peptide backbone.[535] A key feature for the future direction will be structure-guided engineering and/or humanisation to improve affinity and specificity for tumor-restricted STn glycopeptide targets.

4.5. Targeting of the Lewis-Antigens

Besides mAbs to short truncated glycans, mAbs to Lewis antigens have also been exploited. One example is mAbs to the Le^y antigen, strongly upregulated in several cancers, including acute myeloid leukemia and multiple epithelial malignancies.[536−538] Le^y has been targeted using ADCs based on the BR96 antibody, which progressed to phase II trials in advanced NSCLC and metastatic breast cancer.[539,540] Despite initially encouraging results, this program was eventually discontinued because of limited efficacy.[541] More recently, the murine antisialyl-di-Le^a antibody FG129 and its chimeric derivative CH129 have been reported to bind a range of cancers, including colorectal, pancreatic, gastric, and ovarian carcinomas.[86] These antibodies have also been adapted into ADCs that induced cytotoxicity against colorectal and pancreatic cancer cell lines in vitro and suppressed tumor growth in colorectal xenograft models.[86] Le^y has likewise been targeted using CAR T-cell therapy with promising early results. A second-generation CAR based on the humanized Le^y antibody Hu3S193 selectively targeted cancer cell lines *in vitro*, recognizing cells with high or intermediate Le^y expression but not harming cells with low expression, such as neutrophils.[75,542] Furthermore, the mAb was not inhibited by Le^y-positive serum glycoproteins.[75,542] In a phase I trial, four patients with relapsed AML were treated with Le^y CAR T cells without grade 3 or 4 toxicities,[543] and additional phase I studies have evaluated or are evaluating this strategy in advanced solid tumors and lung cancer (NCT03851146 and NCT03198052).

Collectively, these studies demonstrate that both truncated and elongated glycan structures can be exploited as antibody targets in therapeutics, but that particularly Tn-based glycopeptide epitopes are likely to offer a wider therapeutic window because their expression is more restricted to malignant tissue. In this regard, the development of high-affinity antibodies against structurally well-defined truncated glycopeptide epitopes is emerging as a particularly promising strategy.

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4.6. Altering Interactions at the Glycocalyx

Given the prominent and diverse interactions taking place at the glycocalyx, allowing cancer cells to modulate their microenvironment, induce immunosuppression, and mask surface epitopes, many strategies seeking to alter these interactions have emerged. Utilizing glycomimetic compounds, antibodies, and even glycocalyx-modifying enzymes, these therapies generally seek to alter the interactions between the glycocalyx and GBPs, such as galectins, selectins, or siglecs, through competitive or antagonistic interactions or removal of key glycan structures.

Due to the increasingly prominent role of galectins in cancer progression several therapeutic galectin inhibitors have been developed. The first galectin antagonist in clinical trials was the plant derived galactomannan GM-CT-01 (Davanat), developed by Galectin Therapeutics (previously Pro-Pharmaceuticals), which binds to galectin-1 but to different sites than conventional galectin-1 substrates.[544] Davanat entered early clinical testing in patients with various solid tumors alone or in combination with the chemotherapeutic drug 5-fluorouracil (5-FU). It showed preliminary tolerability, but development stalled without convincing confirmatory efficacy or regulatory advancement.[545] In addition, Galectin Therapeutics have pushed the development of galectin-3 inhibitor, Belapectin (GR-MD-02). In murine models, Belapectin has been shown to improve survival with improved immune functions and decreased tumor vasculature.[546] In addition, Belapectin has been tested in a phase 1 trial in combination with pembrolizumab (anti-PD-1) in melanoma, head and neck squamous cell carcinomas, and nonsmall cell lung cancer patients (NCT02575404), and in combination with ipilimumab (anti-CTLA-4) in metastatic melanoma patients (NCT02117362). Belapectin was reported to be well tolerated, with fewer immune-mediated adverse events than pembrolizumAb alone, and significantly increased activation of effector memory T-cells and reduced monocytic MDSCs.[547] Finally, although neutralizing antigalectin-1 antibodies have shown promising preclinical angio- and immunomodulatory activity,[548] galectin-1 antibody therapeutics remain preclinical. By contrast, the unrelated small-molecule inhibitor OTX008 has reached phase I clinical testing in patients with advanced solid tumors, highlighting the potential galectin-1 as an oncology target (NCT01724320).

Similarly, much effort have gone into generating selectin inhibitors for cancer therapeutics. GlycoMimetics has developed the E-selectin inhibitor Uproleselan (GMI-1271), which has completed phase I/II clinical trials in 2018 (NCT02306291) and 2019 (NCT02811822) in patients with acute myeloid leukemia (AML) and multiple myeloma, respectively. Uproleselan treatment was shown to be well tolerated in patients and resulted in high remission rates compared to chemotherapy alone,[549] receiving fast track designation from the FDA. Uproleselan is currently being tested in patients with relapsed or refractory AML in combination with chemotherapy in a phase III clinical trial (NCT03616470). In addition, it is tested in a phase II/III trial and in older AML patients receiving intensive induction chemotherapy (NCT03701308) and for safety in a phase I trial in older or unfit AML patients (NCT04964505). Additionally, GlycoMimetics has designed a novel combined E-selectin and the chemokine receptor CXCR4 antagonist, GMI-1359, which is currently being evaluated for safety in bone metastatic hormone receptor positive breast cancer patients in a phase I clinical trial at Duke University (NCT04197999).

In contrast to Selectins, the most successful therapeutics targeting Siglecs are not glycomimetics, but targeted immunotherapies where Siglecs are used as lineage selective markers in hematological malignancies. This is exemplified by Epratuzumab, Pinatuzumab and Inotuzumab targeting Siglec-2 (CD22) expressing B-cell leukemias,[550−560] and Gemtuzumab targeting Siglec-3 (CD33) expressed in AML.[561−565] However, given the emerging role of Siglecs as functional immune checkpoints there is a growing interest in developing agents that block Siglec signaling. Though few blockers have been generated so far, an encouraging development is the monoclonal Siglec-15 blocking antibody NC318 developed to antagonize interaction with the STn-antigen and thereby suppress tolerance in tumor-infiltrating myeloid cells.[313] A clinical phase I/II trial testing NC318 in patients with advanced solid tumors was recently completed and demonstrated good tolerance as well as single agent activity.[566,567] As a result, NC318 is currently being tested in a phase II clinical trial alone or in combination with pembrolizumab in NSCLC patients, with initial data demonstrating that the combination of NC318 and pembrolizumAb is active in advanced PD-1 axis inhibitor refractory NSCLC (NCT04699123). Alternative strategies to disrupt Siglec interactions by removal of sialoglycan ligands have also been attempted. Small molecule inhibitors mimicking sialic acid precursors have been developed and shown to inhibit endogenous sialylation through competition with CMAS for CMP activation and through competition with the actual sialyltransferases with the CMP-activated inhibitor.[568,569] In murine xenograft models of multiple tumors, including melanoma, intratumor injections with these inhibitors have been shown to reduce tumor growth through enhanced activity and infiltration of NK-cells, $CD8^+$ T cells, while reducing the numbers of regulatory T cells and tolerogenic myeloid immune cells.[570] Together, these findings indicate that local administration and/or activation of such sialyltransferase inhibitors may be a feasible strategy for future cancer therapeutics, especially coupled with other immunotherapies. Additional proof for the importance of sialylation comes from murine studies demonstrating that loss of sialylation capacity restored antitumor immunity and suppressed tumor growth more effectively than anti-PD-L1 therapy in a colorectal cancer model.[571] However, caution should be taken to ensure adverse autoimmunity is not promoted by systemic desialylation, and it is currently unknown whether desialylation could increase MGL-mediated immunosuppression through demasking of STn- to Tn-antigen. Recently, yet another modality to disrupt the Siglec-Sialoglycan pathway has been described. By generating antibody-fusion proteins with glycan binding domains of Siglec-7 and 9, so-called antibody-lectin chimeras (AbLecs), it was demonstrated that tumor targeted lectin decoys displaced endogenous Siglecs and reprogrammed suppressive glycan signaling in cancer.[572]

Similar to small compound sialyltransferase inhibitors, small molecule fucosyltransferase inhibitors, in the form of fucose mimetics, have also been developed. One such molecule (SGN-2FF) has been tested in a phase I clinical trial in patients with advanced solid tumors.[573] SGN-2FF showed promising antitumor effects in mouse models, but while antitumor effects were also observed in human patients the drug induced thromboembolic events that led to the termination of the study.[573] Nevertheless, as with sialyltransferase inhibitors, more site-directed delivery and/or activation of such compounds may make them viable options in the future.

Recently, strategies to directly modify the glycocalyx of cancer cells using sialidases and mucinases have emerged and have

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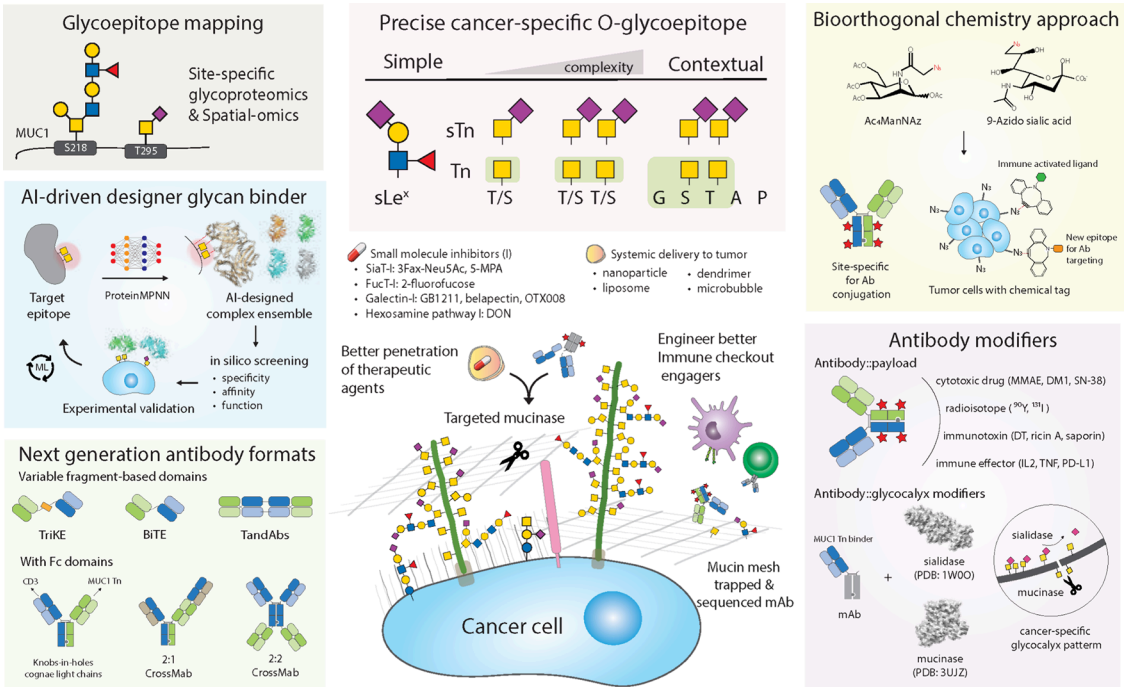


Figure 5. While sLe^x, Tn, and STn are canonical cancer-associated O-glycan epitopes, increasing evidence demonstrates that contextual presentation—in clusters and/or with peptide motifs—further enhances cancer specificity of the binding epitope. (Top left) Site-specific glycoproteomics and spatial-omics enable precise glycoepitope mapping, providing the foundation for rational therapeutic design. (Middle left) AI-driven designer glycan binders leverage protein language models (e.g., ProteinMPNN) to generate optimized paratopes against defined glycoepitopes, followed by ML-based in silico screening for specificity, affinity, and function, and subsequent experimental validation. (Bottom left) Next-generation antibody formats derived from natural IgG architectures include variable fragment-based domains such as bispecific T-cell engagers (BiTEs), trispecific killer engagers (TriKEs), and tandem diabodies (TandAbs), as well as Fc-containing variants (BiTE-Fc, DART-Fc) and knob-in-hole/cognate light chain technologies enabling 2:1 and 2:2 CrossMab configurations for enhanced valency and specificity. (Top right) Bioorthogonal chemistry approaches employ metabolic labeling with unnatural monosaccharides (e.g., Ac_4ManNAz, 9-azido sialic acid) to install chemical handles on tumor cell surfaces, enabling site-specific antibody conjugation, immune-activated ligand targeting, or creation of neo-epitopes for antibody recognition. (Center) Small molecule inhibitors targeting sialyltransferases (3Fax-Neu5Ac, 5-MPA), fucosyltransferases (2-fluorofucose), galectins (GB1211, belapectin, OTX008), and the hexosamine pathway (DON) offer complementary strategies to modulate the tumor glycocalyx. Systemic delivery platforms including nanoparticles, liposomes, dendrimers, and microbubbles facilitate tumor-targeted delivery, while targeted mucinases can improve penetration of therapeutic agents through the mucin mesh. (Right) Antibody modifiers encompass cytotoxic drug conjugates (MMAE, DM1, SN-38), radioisotopes (^{90}Y, ^{131}I), immunotoxins (DT, ricin A, saporin), and immune effectors (IL-2, TNF, PD-L1). (Bottom right) Antibody−glycocalyx modifiers combining MUC1 Tn binders with mucinases (PDB 3UJZ) or sialidases (PDB 1W0O) selectively degrade the cancer-specific glycocalyx, improving target penetration and enabling better immune checkpoint engagement.

interesting implications for potential future therapies. In line with the above desialylation strategies, Bertozzi and colleagues have achieved directed desialylation of cancer cells through conjugation of bacterial sialidases to the anti-HER2 antibody trastuzumAb to direct the sialidase toward HER2-positive tumors. *In vitro*, this construct desialylated cancer cell sialoglycans leading to increased activation of NK cells resulting in improved ADCC.[574] When tested *in vivo* in syngeneic murine breast cancer models, desialylation using this construct resulted in increased immune cell infiltration and improved survival.[575] Palleon pharmaceuticals have tested an Fc-conjugated bisialidase construct (E-602) in several solid tumors in a clinical phase I/II trial (Trial ID NCT05259696) and have other sialidase conjugates in their pipeline, including an anti-PD-L1 conjugated sialidase.[576,577] Similarly, the Bertozzi lab has fused a variant of the bacterial mucinase StcE to a HER2-specific nanobody and used it to selective cleave mucins from cancer cells, in turn making the cancer cells susceptible to NK-directed cytotoxicity and decrease tumor burden in a murine breast cancer model.[577] In parallel, Park et al. have demonstrated that NK-mediated cytotoxicity against cancer cells can be promoted through equipping NK-cells with surface-bound StcE, seemingly through decreasing the thickness of the cancer cell glycocalyx.[578] Recently, a third strategy employing nanovesicles codisplaying StcE and CD47 nanobodies was able to suppress tumorgrowth and metastasis in murine models of colorectal and breast cancer. These nanoparticles were found to selectively degrade cancer cell glycocalyx while also mediating checkpoint blockage, leading to increased M1 macrophage polarization and infiltration of cytotoxic T cells.[579] Though it is unclear whether the effect of these mucinase-strategies that can be attributed to removal of potentially immunosuppressive mucin structures, the unmasking of surface epitopes, or the physical thinning of the glycocalyx, these studies together with the aforementioned sialidase strategies demonstrate that directed modification of the cancer cell glycocalyx could be an emerging strategy to overcome some of the current limitations of cancer immunotherapies.

5. FUTURE DIRECTIONS: TOWARD CONTEXT-DRIVEN EXPLOITATION OF CANCER ASSOCIATED GLYCANS IN PRECISION ONCOLOGY

Taken together, the data reviewed above shows that cancer-associated *O*-GalNAc glycans are context-dependent regulators

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of tumor progression and relevant actionable therapeutic targets. At the same time, the many features that make them unique cancer targets, including their often site-specific and heterogeneous nature as well as their context dependent functions and engagement with glycan-binding proteins, also help explain why many first-generation glycan-targeting strategies achieved limited clinical benefit. The next advance in the field will therefore depend less on identifying glycans in isolation and more on understanding how they are presented in their native molecular and spatial context (Figure 5).

A central unresolved question in this regard is how to define relevant glycoepitopes as they exist in their native context. Which protein sites in the cancer glycocalyx acquire truncated versus extended *O*-glycans, what local constraints determine these outcomes, and how is this regulated? To answer these questions, we must develop better detection reagents and methods, and this will require site-specific characterization that captures both glycan structure and occupancy, likely through integration of glycoproteomics, spatial omics, and measurements on single cells and intact cell surfaces.[580] Ultimately, this will help generate a map of cancer-specific glycopeptide epitopes in their native context. This is essential both for identifying targetable epitopes and for understanding how cancer-associated glycans regulate the binding of glycan-binding proteins involved in metastasis and immune evasion, since recognition occurs on defined carriers, at defined sites, and within dynamic protein scaffolds. In this context, it has been shown that barcoded-lectins can be used to detect binding to accessible glycans on single cells and read in combination with transcriptomes.[428,581−583] This concept is furthermore extended to tissues by Basak, helping to define contextual functions of glycan binding events.[428]

This contextual view is particularly important when considering recognition within the cancer glycocalyx itself. As discussed above, avidity is often more important than affinity in respect to glycan-binding proteins, because many bind only weakly to isolated epitopes but strongly to the same motifs when displayed at high density, for example on extended mucin domains.[425,584] Cancer-associated mucins may therefore drive alternative responses, including tolerogenic signaling, or act as decoys that sequester glycan-binding proteins from their normal sites of action. This concept of mucin glycans as decoys, most likely functioning as a shield and immune checkpoint, has major translational implications for binders to cancer-associated glycans. Here, the tumor glycocalyx is both the target and the decoy, and the dense mucin coat can absorb broadly glycan-reactive binders before they reach actionable cell-surface epitopes. This likely contributes to the limited efficacy of early Tn- and STn-targeted strategies and suggests that the next generation of binders must be designed with greater contextual and topological precision.

Achieving such precision will likely depend on targeting defined glycopeptide antigens as they are presented within the native tumor glycocalyx. To accelerate the development of such binders, several platforms beyond conventional monoclonal antibodies are exploited. These include phage display combined with structure-guided approaches, camelid single-domain antibodies, jawless vertebrate variable lymphocyte receptors, aptamers, engineered lectins and glycoenzymes, as well as synthetic receptors.[585−588] Together, these expand the possibilities for targeting epitopes defined by peptide context, glycan density, linkage, or additional decorations such as sulfation, although they vary in their ability to bridge larger and more complex epitopes containing both peptide and carbohydrate elements. Recent structure-guided phage-display strategies are particularly promising, as fixing glycan recognition in one antibody domain while diversifying the other has enabled the generation of high-affinity binders to Tn- and STn-glycopeptides in context.[588] Nanobodies may offer advantages in tissue penetration, while aptamers and engineered lectins provide complementary formats for imaging and delivery.[586,588]

Once such binders have been generated, the next question is how to best deploy them therapeutically. As we have seen, cancer-selective binders can be functionalized with different modalities, including ADCs, TCBs, CAR-Ts, and radiopharmaceuticals, or combined with glycocalyx- and extracellular matrix-remodelling enzymes.[310,575,577] Especially for ADCs, TCBs, and CAR-Ts, however, a major translational challenge remains the heterogeneous expression of tumor-associated glycans, both within individual tumors and across patients. This is particularly problematic for modalities with limited bystander effects, such as CAR-T cells and T-cell engagers.[589,590] To address this issue, multispecific binders created by combining two glycopeptide-specific binders, or by pairing a glycan-directed binder with a protein-directed binder, for example against EGFR, claudins, EpCAM, or c-MET, may become helpful. Such strategies could broaden tumor coverage, improve internalization, and stabilize immune synapse formation, although the affinity of the protein-directed arm may need to be tuned to avoid healthy tissues.

Beyond improving the binders themselves, a complementary strategy is to manipulate the cancer glycocalyx directly. Metabolic sialylation inhibitors, targeted sialidases, mucin-selective proteases, cathepsin-based debulking strategies, and related glycocalyx-editing approaches may be used either to disrupt glycan-dependent immune suppression or to remove glycan decoys and expose underlying therapeutic targets.[591−593,307,575,577,594] Particularly through Siglecs or MGL, such approaches may also reprogram glyco-immune checkpoints or expose newly actionable surface targets.[595,596,571,597] Particularly in solid tumors, future treatments may therefore combine glycocalyx editing with therapies directed against newly exposed glycoepitopes.

These efforts also intersect naturally with advances in modular coupling of cancer-selective binders to payloads, enzymes, or inhibitors.[598−600] Here, bioorthogonal chemistry enables site-defined conjugation, homogeneous molecules, and tunable release properties.[598,599] Molecules with defined drug-to-antibody ratios and improved stability can be generated through engineered cysteines, aldehyde tags, sortase motifs, or non-canonical amino acids.[598,600] These approaches may be especially valuable when target epitopes reside in the glycocalyx or extracellular matrix rather than on efficiently internalizing receptors. More broadly, click-enabled designs could support local delivery of immune agonists, masked radioligands, glycocalyx-editing enzymes, or extracellular degraders.[577,594] The clinically tested SQ3370 platform already provides proof of principle for *in situ* bioorthogonal activation of a therapeutic payload and illustrates how such strategies could be adapted to cancer-associated *O*-glycan targeting.[601]

Looking further ahead, artificial intelligence is likely to accelerate nearly every step of this pipeline. Generative models, protein language models, structure prediction, and glycopeptide docking are beginning to make it feasible to design antibodies, carbohydrate-binding mimetics, engineered lectins, and other synthetic binders.[602,586,603] AI may also help predict cross-reactivity by integrating glycan-array, tissue-expression, and

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glycoproteomic data, and may eventually enable binders conditioned by tumor-specific features such as pH or protease activity. At present, however, the lack of structural information on glycans in peptide context remains a major limitation, compounded by the large conformational space of glycopeptides. In contrast, AI may more readily support the design of ligands for defined binding pockets in glycan-binding proteins. Most importantly, AI could link target discovery directly to therapeutic design by integrating spatial omics and glycoproteomics to nominate glycopeptide targets and recommend optimal binder architectures. Combined with click chemistry and rapid testing in glyco-engineered cells, tissues, or patient-derived organoids, this could enable iterative active-learning cycles that progressively refine glycan-directed therapeutics.[604]

6. CONCLUSION

Cancer-associated *O*-GalNAc glycans are emerging as more than biomarkers of malignant transformation. Through their natural presentation on proteins, they are context-dependent determinants that can contribute directly to oncogenesis, metastasis, and immune evasion. Truncated *O*-glycans can affect cell−cell and cell-matrix interactions, affect epithelial barriers, destabilize mucin organization, increase susceptibility to proteolysis and inflammation, and directly perturb differentiation and signaling. Both truncated and elongated *O*-glycans also reprogram how tumor cells engage glycan-binding proteins, such as selectins important for metastasis and Siglecs, MGL, and galectins primarily functioning as immune checkpoints. In parallel, site-specific glycosylation by individual GalNAc-transferase isoenzymes is increasingly recognized as a mechanism for regulating selected protein substrates and specific oncogenic pathways. Together, these observations establish the tumor glycocalyx as both a driver to promote cancer growth and a rich source of cancer-selective therapeutic opportunities.

At the same time, the nature of the tumor glycocalyx also poses major obstacles for the efficient treatment of cancer. The expression of cancer associated *O*-glycans is heterogeneous and site-specific. Hence, functional recognition depends on identification of specific epitopes accessible for binding, including carrier context, density, avidity, and secondary modification. Another issue is that although tumor associated glycans represents a source of cancer-selective epitopes, they also function as a decoy on layers of shed mucins preventing access to the tumor surface. These features helped explain why many early glycan-targeting strategies underperformed, but simultaneously direct the development of effective next-generation approaches. Here glycoproteomic and spatial solutions as well as AI-assisted molecular engineering will help select relevant targets that can be used for truly cancer specific targeting matching requirement for specific modalities, including ADCs, TCBs, CAR-Ts, radiopharmaceuticals, or glycocalyx and extracellular matrix-remodeling enzymes.

The broader conclusion is therefore that the future of this field will depend on the precise understanding of how glycans are presented, organized, and detected in living tumors. Achieving this insight will enable us to better exploit cancer-associated O-GalNAc glycans as therapeutic targets and create a powerful example of how molecular complexity of glycans can be translated into cancer specific therapeutic solutions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.chemrev.5c00901.

List of antibodies targeting *O*-glycan TACAs, including antibodies not used in clinical trials, (PDF)

List summarizing studies examining the prevalence of various *O*-glycan TACAs across different types of cancer (XLSX)

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Notes

The authors declare the following competing financial interest(s): H.H.W. is a co-founder and/or consultant and holds equity in EbuMab ApS, Hemab ApS, Cymab ApS, RidgeBio Inc., and GO-Therapeutics Inc. All other authors declare no competing interests.

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Mikkel Koed Møller Aasted, Ph.D., is a former postdoc at the Copenhagen Center for Glycocalyx Research. Through his Ph.D. and postdoctoral research at the Copenhagen Center for Glycocalyx Research, he investigated the expression and regulation of truncated *O*-GalNAc glycans in cancer using genetic engineering, genome-wide CRISPR knockout screening, and organotypic 3D models. Additionally, he has developed immunotherapeutic strategies targeting truncated *O*-GalNAc glycans, including antibodies and CAR T cells, and developed novel multicellular 3D models to assess efficacy and safety of immunotherapeutic modalities *in vitro*.

Thapakorn Jaroentomeechai, Ph.D., is leading the SynGlyco research group at the Department of Biotechnology, Faculty of Science, Mahidol University, and a guest researcher at the Copenhagen Center for

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Glycocalyx Research. He works in the field of synthetic glycobiology, with a focus on engineering bacterial and cell-free systems to produce defined glycoproteins and to interrogate glycan function. He trained at Cornell University with Matthew DeLisa, in collaboration with Michael Jewett, where he copioneered cell-free glycoprotein synthesis (CFGpS), a modular one-pot platform coupling in vitro transcription−translation with bacterial oligosaccharyltransferases to produce N- and O-linked glycoproteins on demand. As an EMBO postdoctoral fellow with Henrik Clausen in Copenhagen, he expanded into mammalian mucin O-glycobiology, characterizing how site-specific O-glycans on mucin tandem repeats shape microbial recognition and epithelial barrier function. His team combines *E. coli* glycoengineering, cell-free synthesis, and glycoproteomics to build programmable pipelines for glycoconjugate vaccines, therapeutic glycoproteins, and diagnostic reagents. He serves on the Editorial Board of *Glycoconjugate Journal*.

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Hiren Joshi is an Associate Professor at the University of Copenhagen and the Danish National Research Foundation Center for Glycocalyx Research. He researches across all aspects of glycobiology, through the lens of data science and informatics. His works include global and systematic analyses of the glycosylation machinery in metazoa, discovery of glycosyltransferase activities in humans, and the mapping of O-GalNAc type glycoproteomes. His group has developed extensive informatic tools to predict and analyse data related to the glycome and glycoproteomes, gaining insight into the functions of individual glycoproteins.

Sean R. Stowell, M.D., Ph.D., is a Professor at Harvard Medical School, Chief of the Division of Transfusion Medicine at Mass General Brigham, and Co-Director of the Harvard Medical School Center for Glycoscience. His laboratory investigates how cell surface glycans regulate immune cell and microbial function, with a particular focus on host−microbe interactions. This work has elucidated key roles for glycans in immune cell development and differentiation, as well as how interactions with distinct microbial glycans shape immune responses. Collectively, these studies aim to define fundamental mechanisms underlying clinically important processes, including transfusion alloimmunity and incompatibility, transplant rejection, susceptibility to infection, tumor immune evasion, and autoimmunity.

Hans H. Wandall, M.D., Ph.D., is Professor of Medical Glycobiology at the University of Copenhagen and Director of the Danish National Research Foundation Center for Glycocalyx Research. He works in the field of glycobiology, with a focus on how glycans regulate cell and tissue biology and how altered glycosylation contributes to disease. His early research helped characterize glycosyltransferases, including egghead and brainiac in arthro-series glycosphingolipid biosynthesis, and the isoform-specific and overlapping functions of polypeptide GalNAc-transferases in mucin-type O-GalNAc glycosylation. His group has developed SimpleCell and CRISPR-engineered tissue systems for functional glycomics and has mapped O-glycoproteomes in cells, tissues, and enveloped viruses. Combining glycoengineering, glycoproteomics, phosphoproteomics, and 3D organotypic models, he has shown how truncated O-glycans can promote epithelial tumorigenesis while creating cancer-specific glycopeptide epitopes. His discoveries have supported glycan-targeted antibody programs and therapeutic translation in cancer and hematology, including through the cofounding of GO Therapeutics and Hemab Therapeutics. He serves on the Editorial Board of Glycobiology.

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■ ABBREVIATIONS

5-FU = 5-fluorouracil
ADC = antibody−drug conjugate
ADCC = antibody-dependent cellular cytotoxicity
ADCP = antibody-dependent cellular phagocytosis
AML = acute myeloid leukemia
APC = antigen-presenting cell
CAR = chimeric antigen receptor
CDC = complement-dependent cytotoxicity
CDX = cell line-derived xenograft
CLR = C-type lectin receptors
CMAS = *N*-acylneuraminate cytidylyltransferase
CMP = cytidine monophosphate
CNS = central nervous system
COSMC = core 1 β3-Gal-T-specific molecular chaperone
COX-2 = cyclooxygenase-2
CPX = cyclophosphamide
CRD = carbohydrate-recognition domain
CRISPR = clustered regularly interspaced short palindromic repeats
DAMP = danger-associated molecular pattern
DAP12 = DNAX activation protein-12
DC = dendritic cell
DC-SIGN = dendritic cell-specific intercellular adhesion molecule-3-grabbing nonintegrin
dST = disialyl Thomsen−Friedenreich antigen
ECM = extracellular matrix
EGF = epidermal growth factor
EGFR = epidermal growth factor receptor
EMT = epithelial-to-mesenchymal transition
ER = endoplasmic reticulum
FACS = fluorescence-activated cell sorting
FDA = Food and Drug Administration
Fuc = fucose
GAG = glycosaminoglycan
Gal = galactose

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GALA = GALNT activation
GalNAc = *N*-acetylgalactosamine
GalNAc-T = polypeptide GalNAc transferase
GBP = glycan binding protein
GEC = gastric epithelial cell
GEF = GTP exchange factor
GI = gastrointestinal
Glc = glucose
GlcA = glucuronic acid
GlcN = glucosamine
GlcNAc = *N*-acetylglucosamine
GP1bα = glycoprotein Ib alpha
GSL = glycosphingolipid
GT = glycosyltransferase
HEV = high endothelial venules
HPA = *Helix pomatia* agglutinin
Hyl = hydroxylysine
IEC = intestinal epithelial cell
IFN-γ = interferon gamma
IGF1R = insulin-like growth factor 1 receptor
IL-2 = interleukin 2
IL-10 = interleukin 10
IL-12 = interleukin 12
IL-15 = interleukin 15
ITAM = immunoreceptor tyrosine-based activation motif
ITIM = immunoreceptor tyrosine-based inhibitory motif
KLH = keyhole limpet hemocyanin
LacNAc = *N*-acetyllactosamine
LDLR = low-density lipoprotein receptor
Le = Lewis
lncRNA = long noncoding RNA
mAb = monoclonal antibody
Man = mannose
MDSC = myeloid-derived suppressor cell
MGL = macrophage galactose-type lectin
miRNA = micro RNA
MMAE = monomethyl auristatin E
MMP-9 = matrix metallopeptidase 9
MMP-14 = matrix metallopeptidase 14
MUC1 = mucin 1
MUC2 = mucin 2
MUC4 = mucin 4
MUC5AC = mucin 5AC
MUC7 = mucin 7
MUC13 = mucin 13
MUC16 = mucin 16
MUC18 = mucin 18
MUC22 = mucin 22
MVA = modified *Vaccinia Ankara*
Neu5Ac = *N*-acetylneuraminic acid
NGS = next-generation sequencing
NHS = *N*-hydroxysuccinimide
NK = natural killer
NMR = nuclear magnetic resonance
NSCLC = nonsmall-cell lung cancer
OSM = ovine submaxillary gland mucin
PAM = palmitic acid
PAMP = pathogen-associated molecular pattern
PDIA4 = protein disulfide isomerase family A member 4
PD-L1 = programmed death-ligand 1
PDPN = podoplanin
PDX = patient-derived xenograft
PGE2 = prostaglandin E2
PTM = posttranslational modification
PNA = peanut agglutinin
PSA = prostate-specific antigen
PSGL-1 = P-selectin glycoprotein ligand-1
RA = radiolabeled antibody
RNA = ribonucleic acid
SAMP = self-associated molecular pattern
scFv = single-chain variable fragment
SCLC = small-cell lung cancer
sgRNA = single guide RNA
SHP-1 = SH2-domain-containing protein tyrosine phosphatase 1
Siglec = sialic acid-binding immunoglobulin-type lectin
sLe = sialyl-Lewis
SPPL3 = signal peptide peptidase-3
ST = sialyl-Thomsen−Friedenreich antigen
STn = sialyl-Thomsen-nouveau antigen
T = Thomsen−Friedenreich antigen
TAA = tumor-associated antigen
TACA = tumor-associated carbohydrate antigen
TAM = tumor-associated macrophage
TCR = T cell receptor
TGF-β = transforming growth factor beta
TIM-3 = T cell immunoglobulin and mucin domain 3 protein
Tn = Thomsen-nouveau antigen
TNF-α = tumor necrosis factor alpha
TR = tandem repeat
Xyl = xylose

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